
CHAPITRE 1

MARTINGALE AND CONTINUOUS SEMIMARTINGALES THEORY

1.1 Filtrations and Processes

1.1.1 Random Processes

Throughout this section, let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let (E, d) be a metric space endowed with its Borel σ -field, and let the index set T be either $\mathbb{N}$ (discrete time) or an interval of $\mathbb{R}$ (continuous time). A stochastic process is a family of E -valued random variables

$$X = (X_t)_{t \in T}.$$

Definition 1.1.1: Stochastic process

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let (E, d) be a metric space endowed with its Borel σ -field, and let T be an index set, typically $T = \mathbb{N}$ (discrete time) or $T \subseteq \mathbb{R}$ (continuous time). An *E-valued stochastic process* indexed by T is a family of random variables

$$X = (X_t)_{t \in T},$$

where each $X_t : \Omega \rightarrow E$ is measurable.

Definition 1.1.2: Sample paths

Let $X = (X_t)_{t \in T}$ be an E -valued stochastic process. For every $\omega \in \Omega$, the mapping

$$T \ni t \mapsto X_t(\omega) \in E$$

is called the *sample path* (or *trajectory*) of X associated with ω .

Definition 1.1.3: Graph of a stochastic process

The *graph* of a sample path of a stochastic process $X = (X_t)_{t \in T}$ corresponding to $\omega \in \Omega$ is the subset of $T \times E$ defined by

$$\text{Graph}(X(\omega)) = \{(t, X_t(\omega)) : t \in T\}.$$

When $E = \mathbb{R}$, this is the usual graph of the function $t \mapsto X_t(\omega)$.

Definition 1.1.4: Modification

Let $X = (X_t)_{t \in T}$ and $\widetilde{X} = (\widetilde{X}_t)_{t \in T}$ be two E -valued stochastic processes defined on the same probability space and indexed by the same set T . We say that $\widetilde{X}$ is a *modification* of X if

$$\forall t \in T, \quad \mathbb{P}(\widetilde{X}_t = X_t) = 1.$$

A modification preserves all finite-dimensional distributions. Consequently, any probabilistic property depending only on finite-dimensional marginals is shared by both processes. In contrast, pathwise properties may differ substantially. For example, when $T = \mathbb{R}_+$ and $E = \mathbb{R}$, a process may admit a modification with continuous sample paths even though its original sample paths are almost surely discontinuous.

Definition 1.1.5: Indistinguishability

The processes X and $\widetilde{X}$ are said to be *indistinguishable* if there exists a measurable set $N \subseteq \Omega$ with $\mathbb{P}(N) = 0$ such that

$$\forall \omega \in \Omega \setminus N, \quad \forall t \in T, \quad X_t(\omega) = \widetilde{X}_t(\omega).$$

Equivalently,

$$\mathbb{P}(\forall t \in T, X_t = \widetilde{X}_t) = 1,$$

keeping in mind that, for an arbitrary index set T , the event above need not be measurable.

Every pair of indistinguishable processes are modifications of one another, but the converse is generally false. Indistinguishable processes coincide almost surely as functions of time, whereas modifications agree only at each fixed time. Throughout these notes, processes are identified up to indistinguishability. Thus, uniqueness statements should always be understood in this sense.

When $T = I \subseteq \mathbb{R}$ is an interval, the distinction disappears under mild regularity assumptions. If both X and $\widetilde{X}$ have continuous sample paths (or merely right-continuous or left-continuous sample paths) outside a $\mathbb{P}$ -null set, then

$$X \text{ is a modification of } \widetilde{X} \iff X \text{ and } \widetilde{X} \text{ are indistinguishable.}$$

Indeed, equality holds almost surely at every rational time by countability, and continuity extends this equality to every $t \in I$.

Theorem 1.1.6: Kolmogorov's Continuity Theorem

Let $I \subseteq \mathbb{R}$ be a bounded interval, and let $X = (X_t)_{t \in I}$ be an E -valued stochastic process, where (E, d) is complete. Suppose there exist constants $q, \varepsilon, C > 0$ such that

$$\mathbb{E}[d(X_s, X_t)^q] \leq C|t - s|^{1+\varepsilon}, \quad s, t \in I.$$

Then X admits a modification $\widetilde{X}$ whose sample paths are Hölder continuous of every order $\alpha \in (0, \varepsilon/q)$. More precisely, for every $\alpha \in (0, \varepsilon/q)$ and every $\omega \in \Omega$, there exists a finite constant $C_\alpha(\omega)$ such that

$$d(\widetilde{X}_s(\omega), \widetilde{X}_t(\omega)) \leq C_\alpha(\omega)|t - s|^\alpha, \quad s, t \in I.$$

In particular, X admits a continuous modification, which is unique up to indistinguishability.

Remarks.

- (i) If I is unbounded, for example $I = \mathbb{R}_+$, the theorem can be applied successively on bounded intervals $[0, 1], [1, 2], \dots$. This yields a modification whose sample paths are locally Hölder continuous of every order $\alpha \in (0, \varepsilon/q)$.
- (ii) It is sufficient to prove the theorem for one fixed exponent $\alpha \in (0, \varepsilon/q)$. Applying the result to a sequence $\alpha_n \uparrow \varepsilon/q$ gives modifications that are necessarily indistinguishable, and therefore coincide almost surely.

Lemma 1.1.7: Dyadic Hölder Lemma

Let $D = \{i2^{-n} : n \geq 1, 0 \leq i < 2^n\} \subset [0, 1)$, and let $f : D \rightarrow E$, where (E, d) is a metric space. Assume that there exist constants $\alpha > 0$ and $K < \infty$ such that

$$d(f((i-1)2^{-n}), f(i2^{-n})) \leq K2^{-n\alpha}$$

for every $n \geq 1$ and every $i = 1, \dots, 2^n - 1$. Then

$$d(f(s), f(t)) \leq \frac{2K}{1 - 2^{-\alpha}}|t - s|^\alpha, \quad s, t \in D.$$

1.1.2 Filtration

Definition 1.1.8: Filtration

A filtration on $(\Omega, \mathcal{A}, \mathbb{P})$ is an increasing family of sub- σ -fields of $\mathcal{A}$. The family in the most applications of this book is indexed by time, which is either discrete ($\mathbb{N}$) or continuous ($\mathbb{R}_+$).

- **The discrete-time case :** A discrete-time filtration is a family $(\mathcal{F}_n)_{n \in \mathbb{N}}$ of σ -fields such that :

$$\mathcal{F}_0 \subset \mathcal{F}_1 \subset \mathcal{F}_2 \subset \cdots \subset \mathcal{A}$$

- **The continuous-time case :** A continuous-time filtration is a family $(\mathcal{F}_t)_{t \in \mathbb{R}_+}$ of σ -fields such that :

$$\mathcal{F}_0 \subset \mathcal{F}_s \subset \mathcal{F}_t \subset \mathcal{F}$$

We also say that $(\Omega, \mathcal{A}, (\mathcal{F}_n)_{n \in \mathbb{N}}, \mathbb{P})$ and $(\Omega, \mathcal{A}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ are filtered probability spaces.

The σ -field $\mathcal{F}_n$ (resp. $\mathcal{F}_t$) gathers the information available at time n (resp. t), events that are $\mathcal{F}_n$ -measurable (resp. $\mathcal{F}_t$ -measurable) are interpreted as those that depend only on what has happened up to time n (resp. t). We will write :

$$\mathcal{F}_\infty = \begin{cases} \bigvee_{n \in \mathbb{N}} \mathcal{F}_n := \sigma(\bigcup_{n \in \mathbb{N}} \mathcal{F}_n) & \text{in the discrete-time case} \\ \bigvee_{t \geq 0} \mathcal{F}_t := \sigma(\bigcup_{t \geq 0} \mathcal{F}_t) & \text{in the continuous-time case} \end{cases}$$

If $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) is a random process, then, for every $n \in \mathbb{N}$ (resp. $t \geq 0$), we may define $\mathcal{F}_n^X$ (resp. $\mathcal{F}_t^X$) as the smallest σ -field on Ω for which $X_0, X_1, \dots, X_n$ (resp. $X_s \quad s \leq t$) are measurables. In other words, we have :

$$\begin{cases} \mathcal{F}_n^X = \sigma(X_0, X_1, \dots, X_n) & \text{in the discrete-time case} \\ \mathcal{F}_t^X = \sigma(X_s : s \leq t) & \text{in the continuous-time case} \end{cases}$$

Then $(\mathcal{F}_n^X)_{n \in \mathbb{N}}$ (resp. $(\mathcal{F}_t^X)_{t \geq 0}$) is a filtration called *the canonical filtration* of $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$).

Remark : Let $(\mathcal{F}_t)_{0 \leq t \leq \infty}$ be a filtration on $(\Omega, \mathcal{F}, P)$. We set, for every $t \geq 0$

$$\mathcal{F}_{t+} = \bigcap_{s > t} \mathcal{F}_s,$$

and $\mathcal{F}_{\infty+} = \mathcal{F}_\infty$. Note that $\mathcal{F}_t \subset \mathcal{F}_{t+}$ for every $t \in [0, \infty]$. The collection $(\mathcal{F}_{t+})_{0 \leq t \leq \infty}$ is also a filtration. We say that the filtration $(\mathcal{F}_t)$ is right-continuous if

$$\mathcal{F}_{t+} = \mathcal{F}_t, \quad \forall t \geq 0.$$

Let $(\mathcal{F}_t)$ be a filtration and let $\mathcal{N}$ be the class of all $(\mathcal{F}_\infty, P)$ -negligible sets (i.e. $A \in \mathcal{N}$ if there exists an $A' \in \mathcal{F}_\infty$ such $A \subset A'$ and $P(A') = 0$). The filtration is said to be

complete if $\mathcal{N} \subset \mathcal{F}_0$ (and thus $\mathcal{N} \subset \mathcal{F}_t$ for every t).

If $(\mathcal{F}_t)$ is not complete, it can be completed by setting $\mathcal{F}'_t = \mathcal{F}_t \vee \sigma(\mathcal{N})$, for every $t \in [0, \infty]$, using the notation $\mathcal{F}_t \vee \sigma(\mathcal{N})$ for the smallest σ -field that contains both $\mathcal{F}_t$ and $\sigma(\mathcal{N})$ (recall that $\sigma(\mathcal{N})$ is the σ -field generated by $\mathcal{N}$). We will often apply this completion procedure to the canonical filtration of a random process $(X_t)_{t \geq 0}$, and call the resulting filtration the *completed canonical filtration* of X .

Assumption : *In this book we will consider only filtrations that are right-continuous and complete.*

1.1.3 Measurability of processes

For the discrete-time case, we have only one definition for measurability with respect to a filtration, but for the continuous-time case, we have 3 types/definitions of measurability. We start with the discrete-time case.

Definition 1.1.9: Adapted Process [discrete]

A random process $(X_n)_{n \in \mathbb{N}}$ is said to be adapted to the filtration $(\mathcal{F}_n)_{n \in \mathbb{N}}$ if, for every $n \in \mathbb{N}$, X_n is $\mathcal{F}_n$ -measurable.

In the continuous-time case, we have the following definitions.

Definition 1.1.10: Measurable Process [continuous]

A process $X = (X_t)_{t \geq 0}$ with values in a measurable space $(E, \mathcal{E})$ is said to be measurable if the mapping

$$(\omega, t) \mapsto X_t(\omega)$$

defined on $\Omega \times \mathbb{R}_+$ equipped with the product σ -field $\mathcal{F} \otimes \mathcal{B}(\mathbb{R}_+)$ is measurable. (We recall that $\mathcal{B}(\mathbb{R}_+)$ stands for the Borel σ -field of $\mathbb{R}_+$.)

This is stronger than saying that, for every $t \geq 0$, X_t is $\mathcal{F}$ -measurable. On the other hand, considering for instance the case where $E = \mathbb{R}$, it is easy to see that if the sample paths of X are continuous, or only right-continuous, the fact that X_t is $\mathcal{F}$ -measurable for every t implies that the process is measurable in the previous sense – see the argument in the proof of Proposition 1.1.12 below.

Definition 1.1.11: Adapted and Progressive Process

A random process $(X_t)_{t \geq 0}$ with values in a measurable space $(E, \mathcal{E})$ is called **adapted** if, for every $t \geq 0$, X_t is $\mathcal{F}_t$ -measurable. This process is said to be **progressive** if, for every $t \geq 0$, the mapping

$$(\omega, s) \mapsto X_s(\omega)$$

defined on $\Omega \times [0, t]$ is measurable for the σ -field $\mathcal{F}_t \otimes \mathcal{B}([0, t])$.

Proposition 1.1.12

Let $(X_t)_{t \geq 0}$ be a random process with values in a metric space (E, d) (equipped with its Borel σ -field). Suppose that X is adapted and that the sample paths of X are right-continuous (i.e. for every $\omega \in \Omega$, $t \mapsto X_t(\omega)$ is right-continuous). Then X is progressive. The same conclusion holds if one replaces right-continuous by left-continuous.

Proof for Proposition.

We treat only the case of right-continuous sample paths, as the other case is similar. Fix $t > 0$. For every $n \geq 1$ and $s \in [0, t]$, define a random variable X_s^n by setting

$$X_s^n = X_{kt/n} \quad \text{if } s \in \left[(k-1)\frac{t}{n}, k\frac{t}{n} \right), k \in \{1, \dots, n\},$$

and $X_t^n = X_t$. The right-continuity of sample paths ensures that, for every $s \in [0, t]$ and $\omega \in \Omega$,

$$X_s(\omega) = \lim_{n \rightarrow \infty} X_s^n(\omega).$$

On the other hand, for every Borel subset A of E ,

$$\begin{aligned} \{(\omega, s) \in \Omega \times [0, t] : X_s^n(\omega) \in A\} &= (\{X_t \in A\} \times \{t\}) \\ &\cup \left(\bigcup_{k=1}^n \left(\{X_{kt/n} \in A\} \times \left[\frac{(k-1)t}{n}, \frac{kt}{n} \right) \right) \right) \end{aligned}$$

which belongs to the σ -field $\mathcal{F}_t \otimes \mathcal{B}([0, t])$. Hence, for every $n \geq 1$, the mapping $(\omega, s) \mapsto X_s^n(\omega)$, defined on $\Omega \times [0, t]$, is measurable for $\mathcal{F}_t \otimes \mathcal{B}([0, t])$. Since a pointwise limit of measurable functions is also measurable, the same measurability property holds for the mapping $(\omega, s) \mapsto X_s(\omega)$ defined on $\Omega \times [0, t]$. It follows that the process X is progressive. ■

The progressive σ -field. The collection $\mathcal{P}$ of all sets $A \in \mathcal{F} \otimes \mathcal{B}(\mathbb{R}_+)$ such that the process

$$X_t(\omega) = \mathbf{1}_A(\omega, t)$$

is progressive forms a σ -field on $\Omega \times \mathbb{R}_+$, which is called the progressive σ -field. A subset A of $\Omega \times \mathbb{R}_+$ belongs to $\mathcal{P}$ if and only if, for every $t \geq 0$,

$$A \cap (\Omega \times [0, t]) \in \mathcal{F}_t \otimes \mathcal{B}([0, t]).$$

One then verifies that a process X is progressive if and only if the mapping

$$(\omega, t) \mapsto X_t(\omega)$$

is measurable on $\Omega \times \mathbb{R}_+$ equipped with the σ -field $\mathcal{P}$.

1.2 Martingales and stopping times

1.2.1 Martingales

Definition 1.2.1: Martingale

An adapted real-valued process $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) such that $X_n \in L^1$ for every $n \in \mathbb{N}$ (resp. $X_t \in L^1$ for every $t \geq 0$) is called

- a **martingale** if :

$$\begin{cases} \forall n \in \mathbb{N} & X_n = \mathbb{E}[X_{n+1}|F_n] & \text{For the discret time} \\ \forall s \geq t \geq 0 & X_t = \mathbb{E}[X_s|F_t] & \text{For the continuous time} \end{cases}$$

- a **supermartingale** if :

$$\begin{cases} \forall n \in \mathbb{N} & X_n \geq \mathbb{E}[X_{n+1}|F_n] & \text{For the discret time} \\ \forall s \geq t \geq 0 & X_t \geq \mathbb{E}[X_s|F_t] & \text{For the continuous time} \end{cases}$$

- a **submartingale** if :

$$\begin{cases} \forall n \in \mathbb{N} & X_n \leq \mathbb{E}[X_{n+1}|F_n] & \text{For the discret time} \\ \forall s \geq t \geq 0 & X_t \leq \mathbb{E}[X_s|F_t] & \text{For the continuous time} \end{cases}$$

If $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) is a submartingale, $(-X_n)_{n \in \mathbb{N}}$ (resp. $(-X_t)_{t \geq 0}$) is a supermartingale. For this reason, some of the results below are stated for supermartingales only, but the analogous results for submartingales immediately follow.

Proposition 1.2.2

Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}_+$ be a convex function, and let $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) be an adapted process, such that $\mathbb{E}[\varphi(X_n)] < \infty$ for every $n \in \mathbb{N}$ (resp. $\mathbb{E}[\varphi(X_t)] < \infty$ for every $t \geq 0$).

- If $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) is a martingale, then $(\varphi(X_n))_{n \in \mathbb{N}}$ (resp. $(\varphi(X_t))_{t \geq 0}$) is a submartingale.
- If $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) is a submartingale, and if we assume in addition that φ is increasing, $(\varphi(X_n))_{n \in \mathbb{N}}$ (resp. $(\varphi(X_t))_{t \geq 0}$) is a submartingale.

In particular, if X_n (resp. X_t) is a martingale, $|X_n|$ (resp. $|X_t|$) is a submartingale (and so is X_n^2 , provided that $\mathbb{E}[X_n^2] < \infty$ for every n) and, if X_n (resp. X_t) is a submartingale, $(X_n)^+$ (resp. $(X_t)^+$) is also a submartingale.

Proposition 1.2.3

Let $(M_n)_{n \in \mathbb{N}}$ (resp. $(M_t)_{t \geq 0}$) be a square integrable martingale ($\in L^2$). Then :

- **Discrete-time case** : If $n \geq m \geq 1$. Then,

$$\mathbb{E} \left[\sum_{k=m}^{n-1} (M_{k+1} - M_k)^2 \mid \mathcal{F}_m \right] = \mathbb{E}[M_n^2 - M_m^2 \mid \mathcal{F}_m] = \mathbb{E}[(M_n - M_m)^2 \mid \mathcal{F}_m].$$

In particular,

$$\mathbb{E} \left[\sum_{i=m}^{n-1} (M_{k+1} - M_{t_k})^2 \right] = \mathbb{E}[M_n^2 - M_m^2] = \mathbb{E}[(M_n - M_m)^2].$$

- **Continuous-time case** : If $0 \leq s < t$ and let $s = t_0 < t_1 < \dots < t_p = t$ be a subdivision of the interval $[s, t]$. Then,

$$\mathbb{E} \left[\sum_{i=1}^p (M_{t_i} - M_{t_{i-1}})^2 \mid \mathcal{F}_s \right] = \mathbb{E}[M_t^2 - M_s^2 \mid \mathcal{F}_s] = \mathbb{E}[(M_t - M_s)^2 \mid \mathcal{F}_s].$$

In particular,

$$\mathbb{E} \left[\sum_{i=1}^p (M_{t_i} - M_{t_{i-1}})^2 \right] = \mathbb{E}[M_t^2 - M_s^2] = \mathbb{E}[(M_t - M_s)^2].$$

1.2.2 Stopping times**Definition 1.2.4: Stopping time**

A random variable $T : \Omega \longrightarrow \mathbb{N} \cup \{+\infty\}$ (resp. $T : \Omega \longrightarrow [0, \infty]$) is a **stopping time** of the filtration $(\mathcal{F}_n)$ (resp. $(\mathcal{F}_t)$) if $\{T \leq n\} \in \mathcal{F}_n \quad \forall n \in \mathbb{N}$ (resp. $\{T \leq t\} \in \mathcal{F}_t \quad \forall t \geq 0$). The σ -field of the past before T is then defined by

$$\begin{cases} \mathcal{F}_T = \{A \in \mathcal{F}_\infty : \forall n \in \mathbb{N}, A \cap \{T \leq n\} \in \mathcal{F}_n\}. & \text{For the discrete time} \\ \mathcal{F}_T = \{A \in \mathcal{F}_\infty : \forall t \geq 0, A \cap \{T \leq t\} \in \mathcal{F}_t\}. & \text{For the continuous time} \end{cases}$$

The reader will verify that $\mathcal{F}_T$ is indeed a σ -field.

In what follows, discrete (resp. continuous) "stopping time" will mean stopping time of the filtration $(\mathcal{F}_n)$ (resp. $(\mathcal{F}_t)$) unless otherwise specified. If T is a continuous stopping time, we also have $\{T < t\} \in \mathcal{F}_t$ for every $t > 0$, by considering $\{T < t\} = \bigcup_{n=1}^{\infty} \left\{T \leq t - \frac{1}{n}\right\}$, and moreover

$$\{T = \infty\} = \left(\bigcup_{n=1}^{\infty} \{T \leq n\} \right)^c \in \mathcal{F}_\infty.$$

A good advantage of assuming the right continuity of the filtration is that, in this case, the definition of stopping times in the continuous time case can be equivalently stated by replacing $\{T \leq t\} \in \mathcal{F}_t$ by $\{T < t\} \in \mathcal{F}_t$ for every $t \geq 0$. In fact :

$$\{T \leq t\} = \bigcap_{n=1}^{\infty} \left\{ T < t + \frac{1}{n} \right\} \in \mathcal{F}_{t+} = \mathcal{F}_t$$

Proposition 1.2.5

Let S and T be two stopping times

- If $S \leq T$. Then, $\mathcal{F}_S \subset \mathcal{F}_T$.
- We have in general, $S \wedge T$ and $S \vee T$ are also stopping times. Moreover $\mathcal{F}_{S \wedge T} = \mathcal{F}_S \cap \mathcal{F}_T$.

Proof for Proposition.

Can be easily proven by using the definition of stopping times and σ -field of stopping times and manipulation of sets. ■

Lemma 1.2.6: Measurability w.r.t. stopped filtration

Let T be a stopping time. A function $\omega \mapsto Y(\omega)$ defined on the set $\{T < \infty\}$ and taking values in the measurable set $(E, \mathcal{E})$ is $\mathcal{F}_T$ -measurable if and only if, for every $n \in \mathbb{N}$ (resp. $t \geq 0$), the restriction of Y to the set $\{T = n\}$ is $\mathcal{F}_n$ measurable (resp. $\{T \leq t\}$ $\mathcal{F}_t$ -measurable).

Proof for Lemma

For the discrete-time case, first assume that, for every $n \in \mathbb{N}$, the restriction of Y to $\{T = n\}$ is $\mathcal{F}_n$ -measurable. Then, for every measurable subset A of E ,

$$\{Y \in A\} \cap \{T \leq n\} = \bigcup_{k=0}^n (\{Y \in A\} \cap \{T = k\}) \in \mathcal{F}_n.$$

Letting $n \rightarrow \infty$, we first obtain that $\{Y \in A\} \in \mathcal{F}_\infty$, and then we deduce from the previous display that $\{Y \in A\} \in \mathcal{F}_T$. Conversely, if Y is $\mathcal{F}_T$ -measurable, $\{Y \in A\} \in \mathcal{F}_T$ and thus $\{Y \in A\} \cap \{T = n\} = \{Y \in A\} \cap \{T \leq n\} \setminus \{Y \in A\} \cap \{T \leq n-1\} \in \mathcal{F}_n$, giving the desired result.

For the continuous-time case, first assume that, for every $t \geq 0$, the restriction of Y to $\{T \leq t\}$ is $\mathcal{F}_t$ -measurable. Then, for every measurable subset A of E ,

$$\{Y \in A\} \cap \{T \leq t\} \in \mathcal{F}_t.$$

Letting $t \rightarrow \infty$, we first obtain that $\{Y \in A\} \in \mathcal{F}_\infty$, and then we deduce from the previous display that $\{Y \in A\} \in \mathcal{F}_T$.

Conversely, if Y is $\mathcal{F}_T$ -measurable, $\{Y \in A\} \in \mathcal{F}_T$ and thus $\{Y \in A\} \cap \{T \leq t\} \in \mathcal{F}_t$,

giving the desired result. ■

Theorem 1.2.7: Stopped process Measurability

Let $(Y_n)_{n \in \mathbb{N}}$ (resp. $(Y_t)_{t \geq 0}$) be an adapted (resp. a progressive) process, and let T be a stopping time. Then the random variable Y_T , which is defined on the event $\{T < \infty\}$ by $Y_T(\omega) = Y_{T(\omega)}(\omega)$, is $\mathcal{F}_T$ -measurable.

Proof for Theorem.

Let B be a Borel subset of $\mathbb{R}$. Let's begin with the discrete-time case, consider the restriction of Y_T to the set $\{T = n\}$, which is given by Y_n on this set. Since Y_n is $\mathcal{F}_n$ -measurable, the restriction of Y_T to $\{T = n\}$ is also $\mathcal{F}_n$ -measurable. Hence, by Lemma 1.2.6, Y_T is $\mathcal{F}_T$ -measurable. For the continuous-time case. Let $t \geq 0$. The restriction to $\{T \leq t\}$ of the function $\omega \mapsto X_T(\omega)$ is the composition of the two mappings

$$\begin{aligned} \{T \leq t\} \ni \omega &\mapsto (\omega, T(\omega) \wedge t) \\ &\mathcal{F}_t \quad \mathcal{F}_t \otimes \mathcal{B}([0, t]) \end{aligned}$$

and

$$\begin{aligned} \Omega \times [0, t] \ni (\omega, s) &\mapsto X_s(\omega) \\ &\mathcal{F}_t \otimes \mathcal{B}([0, t]) \quad \mathcal{E} \end{aligned}$$

which are both measurable (the first one since $T \wedge t$ is $\mathcal{F}_t$ -measurable, by Proposition 1.2.5, and the second one by the definition of a progressive process). It follows that the restriction to $\{T \leq t\}$ of the function $\omega \mapsto X_T(\omega)$ is $\mathcal{F}_t$ -measurable, which gives the desired result by the lemma 1.2.6. ■

If T is a stopping time, then, for every $n \in \mathbb{N}$ (resp. $t \geq 0$), $n \wedge T$ (resp. $t \wedge T$) is also a stopping time (Proposition 1.2.5) and it follows from the preceding theorem that $Y_{n \wedge T}$ (resp. $Y_{t \wedge T}$) is $\mathcal{F}_{n \wedge T}$ -measurable (resp. $\mathcal{F}_{t \wedge T}$ -measurable), hence also $\mathcal{F}_n$ -measurable (resp. $\mathcal{F}_t$ -measurable) by Proposition 1.2.5.

Proposition 1.2.8

If (S_n) is a sequence of stopping times then :

- if (S_n) is monotone increasing, then $S = \lim \uparrow S_n$ is a stopping time.
- If (S_n) is monotone decreasing, then $S = \lim \downarrow S_n$ is a stopping time and

$$\mathcal{F}_S = \bigcap_n \mathcal{F}_{S_n}$$

Proof for Proposition.

- If (S_n) is monotone increasing, then for every $t \geq 0$, $\{S \leq t\} = \bigcap_{n=1}^{\infty} \{S_n \leq t\}$, which belongs to the σ -field $\mathcal{F}_t$ since S_n is a stopping time for every n . Hence, S is a stopping time. same argument for the discrete time case.
- If (S_n) is monotone decreasing, then for every $t \geq 0$, $\{S < t\} = \bigcup_{n=1}^{\infty} \{S_n < t\}$,

which belongs to the σ -field $\mathcal{F}_t$ since S_n is a stopping time for every n . Hence, S is a stopping time by the fact that $\mathcal{F}_t$ is right-continuous [check remarks after the definition 1.2.4].

- For the last assertion we first verify that $\mathcal{F}_S \subset \bigcap_n \mathcal{F}_{S_n}$ thanks to last proposition 1.2.5. Conversely, if $A \in \bigcap_n \mathcal{F}_{S_n}$, then

$$A \cap \{S < t\} = \bigcup_{n=1}^{\infty} A \cap \{S_n < t\} \in \mathcal{F}_t,$$

which gives $A \in \mathcal{F}_S$ again this is allowed thanks to the right-continuity of the filtration. ■

Proposition 1.2.9

Let T be a stopping time and let S be an $\mathcal{F}_T$ -measurable random variable with values in $[0, \infty]$, such that $S \geq T$. Then S is also a stopping time. In particular, if T is a continuous stopping time,

$$T_n = \frac{\lceil 2^n T \rceil}{2^n} \cdot \mathbb{1}_{\{T < \infty\}} + \infty \cdot \mathbb{1}_{\{T = \infty\}}, \quad n = 0, 1, 2, \dots$$

defines a sequence of stopping times that decreases to T .

Proof for Proposition.

For the first assertion, we write, for every $n \in \mathbb{N}$,

$$\{S \leq n\} = \{S \leq n\} \cap \{T \leq n\} \in \mathcal{F}_n$$

since $\{S \leq n\}$ is $\mathcal{F}_T$ -measurable. And similar argument holds for the continuous time case. Further more, the second assertion follows since $T_n \geq T$, and T_n is a function of T , hence $\mathcal{F}_T$ -measurable, and $T_n \downarrow T$ as $n \uparrow \infty$ by construction. ■

Finally lets defining hitting time.

Definition 1.2.10: Hitting time

Let $(X_t)_{t \geq 0}$ be an adapted process with values in a metric space (E, d) .

1. Assume that the sample paths of X are right-continuous, and let O be an open subset of E . Then

$$T_O = \inf\{t \geq 0 : X_t \in O\}$$

is a stopping time of the filtration $(\mathcal{F}_t)$.

2. Assume that the sample paths of X are continuous, and let F be a closed

subset of E . Then

$$T_F = \inf\{t \geq 0 : X_t \in F\}$$

is a stopping time (of the filtration $(\mathcal{F}_t)$).

In fact the stopping time property can be proven by considering the two sets :

$$\begin{cases} \{T_O < t\} &= \bigcup_{s \in [0, t) \cap \mathbb{Q}} \{X_s \in O\} \in \mathcal{F}_t, \\ \{T_F \leq t\} &= \left\{ \inf_{0 \leq s \leq t} d(X_s, F) = 0 \right\} = \left\{ \inf_{s \in [0, t] \cap \mathbb{Q}} d(X_s, F) = 0 \right\} \in \mathcal{F}_t. \end{cases}$$

1.2.3 Two Important Lemmas

Lemma 1.2.11: Optional stopping theorem, first version

Suppose that $(X_n)_{n \in \mathbb{N}}$ is a martingale (resp. a supermartingale) and let T be a stopping time. Then $(X_{n \wedge T})_{n \in \mathbb{N}}$ is also a martingale (resp. a supermartingale). In particular, if the stopping time T is bounded, one has $X_T \in L^1$, and

$$\mathbb{E}[X_T] = \mathbb{E}[X_0] \quad (\text{resp. } \mathbb{E}[X_T] \leq \mathbb{E}[X_0]).$$

Proof for Lemma

Fix $n \in \mathbb{N}$. We wish to show that

$$\mathbb{E}[X_{(n+1) \wedge T} \mid \mathcal{F}_n] = X_{n \wedge T}.$$

Observe the pointwise decomposition

$$X_{(n+1) \wedge T} - X_{n \wedge T} = (X_{n+1} - X_n) \mathbb{1}_{T > n},$$

Therefore it suffices to show

$$\mathbb{E}[(X_{n+1} - X_n), \mathbb{1}_{T > n} \mid \mathcal{F}_n] = 0.$$

Since T is a stopping time, the event $\{T > n\} = \{T \leq n\}^c$ belongs to $\mathcal{F}_n$, so the indicator $\mathbb{1}_{\{T > n\}}$ is $\mathcal{F}_n$ -measurable. Taking it outside the conditional expectation and using the martingale property of (X_n) :

$$\begin{aligned} \mathbb{E}[(X_{n+1} - X_n), \mathbb{1}_{\{T > n\}} \mid \mathcal{F}_n] &= \mathbb{1}_{\{T > n\}} \mathbb{E}[X_{n+1} - X_n \mid \mathcal{F}_n] \\ &= \mathbb{1}_{\{T > n\}} \cdot 0 \\ &= 0. \end{aligned}$$

Hence $(X_{n \wedge T})_{n \in \mathbb{N}}$ is a martingale.

If the stopping time T is bounded above by the constant N , $X_T = X_{N \wedge T}$ is in L^1 ,

and $\mathbb{E}[X_T] = \mathbb{E}[X_{N \wedge T}] = \mathbb{E}[X_0]$. For the supermartingale case just replace equalities with inequalities. ■

Lemma 1.2.12

Let $(X_n)_{n \in \mathbb{N}}$ be a supermartingale, and let S and T be two bounded stopping times such that $S \leq T$. Then

$$\mathbb{E}[X_T] \leq \mathbb{E}[X_S].$$

Proof for Lemma

We began with the discrete-time case. Let $(X_n, \mathcal{F}_n)_{n \geq 0}$ be a supermartingale and let $S \leq T \leq N$ be bounded stopping times. Define the stopped process increments :

$$X_T - X_S = \sum_{k=1}^N \mathbf{1}_{S < k \leq T} (X_k - X_{k-1}).$$

Take expectation :

$$\mathbb{E}[X_T - X_S] = \sum_{k=1}^N \mathbb{E} [\mathbf{1}_{S < k \leq T} (X_k - X_{k-1})].$$

Since S, T are stopping times, the event $\{S < k \leq T\} \in \mathcal{F}_{k-1}$. Hence the indicator is $\mathcal{F}_{k-1}$ -measurable. Using the supermartingale property, $\mathbb{E}[X_k - X_{k-1} \mid \mathcal{F}_{k-1}] \leq 0$, we get

$$\mathbb{E} [\mathbf{1}_{S < k \leq T} (X_k - X_{k-1})] \leq 0.$$

Summing over k , and using the fact that $X_T, X_S \in L^1$ (by Lemma 1.2.11),

$$\mathbb{E}[X_T - X_S] \leq 0, \quad \Rightarrow \quad \mathbb{E}[X_T] \leq \mathbb{E}[X_S].$$

1.3 Martingale convergence theory

1.3.1 Upcrossing numbers

Discret time

Consider first a deterministic real sequence $y = (y_n)_{n \in \mathbb{N}}$, and $a < b$, the **upcrossing number** of this sequence along $[a, b]$ before time n , denoted by $M_{ab}^y(n)$, is the largest integer k such that there exists a strictly increasing finite sequence

$$m_1 < n_1 < m_2 < n_2 < \cdots < m_k < n_k$$

of nonnegative integers smaller than or equal to n with the properties $y_{m_i} \leq a$ and $y_{n_i} \geq b$, for every $i \in \{1, \dots, k\}$. In what follows we consider a sequence $Y = (Y_n)_{n \in \mathbb{N}}$ of real random variables, and the associated upcrossing number $M_{ab}^Y(n)$ is then an integer-valued

random variable. where $M_{ab}^y(0) = 0$ if such a sequence does not exist. We will use the following simple analytic lemma, whose proof is left to the reader.

Lemma 1.3.1

The sequence $(y_n)_{n \in \mathbb{N}}$ converges in $\overline{\mathbb{R}}$ if and only if, for every choice of the rationals a and b with $a < b$, we have $M_{ab}^y(n) < \infty$.

Lemma 1.3.2: Doob's Upcrossing Inequality

Let $X = (X_n)_{n \in \mathbb{N}}$ be a supermartingale. Then, for every reals $a < b$ and every $n \in \mathbb{N}$,

$$E[M_{ab}^Y(n)] \leq \frac{1}{b-a} E[(Y_n - a)^-]$$

Continuous time

Let $f : I \rightarrow \mathbb{R}$ be a function defined on a subset I of $\mathbb{R}_+$. If $a < b$, the upcrossing number of f along $[a, b]$, denoted by $M_{ab}^f(I)$, is the maximal integer $k \geq 1$ such that there exists a finite increasing sequence $s_1 < t_1 < \dots < s_k < t_k$ of elements of I such that $f(s_i) \leq a$ and $f(t_i) \geq b$ for every $i \in \{1, \dots, k\}$ (if, even for $k = 1$, there is no such subsequence, we take $M_{ab}^f(I) = 0$, and if such a subsequence exists for every $k \geq 1$, we take $M_{ab}^f(I) = \infty$). Upcrossing numbers are a convenient tool to study the regularity of functions.

In the next lemma, the notation

$$\lim_{s \downarrow t} f(s) \quad (\text{resp. } \lim_{s \uparrow t} f(s))$$

means

$$\lim_{s \downarrow t, s > t} f(s) \quad (\text{resp. } \lim_{s \uparrow t, s < t} f(s)).$$

We say that $g : \mathbb{R}_+ \rightarrow \mathbb{R}$ is **càdlàg** (for the French "continue à droite avec des limites à gauche") if g is right-continuous and has left-limits at every $t > 0$.

Lemma 1.3.3

Let D be a countable dense subset of $\mathbb{R}_+$ and let f be a real function defined on D . We assume that, for every $T \in D$,

- the function f is bounded on $D \cap [0, T]$;
- for all rationals a and b such that $a < b$,

$$M_{ab}^f(D \cap [0, T]) < \infty.$$

Then, the right-limit

$$f(t+) := \lim_{s \downarrow t, s \in D} f(s)$$

exists for every real $t \geq 0$, and similarly the left-limit

$$f(t-) := \lim_{s \uparrow t, s \in D} f(s)$$

exists for every real $t > 0$. Furthermore, the function $g : \mathbb{R}_+ \rightarrow \mathbb{R}$ defined by $g(t) = f(t+)$ is càdlàg.

We omit the proof of this analytic lemma. It is important to note that the right and left-limits $f(t+)$ and $f(t-)$ are defined for every $t \geq 0$ ($t > 0$ in the case of $f(t-)$) and not only for $t \in D$.

Lemma 1.3.4: Doob's Upcrossing Inequality

Let D be a countable dense subset of $\mathbb{R}_+$ and let $X = (X_t)_{t \geq 0}$ be a supermartingale. Then, for every reals $a < b$ and every $t \geq 0$,

$$\mathbb{E}[M_{ab}^X(D \cap [0, T])] \leq \frac{1}{b-a} \mathbb{E}[(X_T - a)^-]$$

Can be obtained by monotone convergence applied on the discrete version Lemma 1.3.2 by considering a sequence of finite subsets of D .

1.3.2 L^p Convergence and Doob's inequalities

Theorem 1.3.5: Doob's supermartingale inequalities

1. (Maximal inequality for submartingales) Let $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) be a submartingale (with right-continuous sample paths in the continuous case). Then, for every $n \in \mathbb{N}$ (resp. $t > 0$) and every $\lambda > 0$,

$$\begin{cases} \lambda \mathbb{P} \left(\sup_{0 \leq m \leq n} |X_m| > \lambda \right) \leq \mathbb{E}[X_n^+] & \text{For the discrete case} \\ \lambda \mathbb{P} \left(\sup_{0 \leq s \leq t} |X_s| > \lambda \right) \leq \mathbb{E}[X_t^+] & \text{For the continuous case} \end{cases}$$

2. (Maximal inequality for supermartingales) Let $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) be a supermartingale (with right-continuous sample paths in the continuous case). Then, for every $n \in \mathbb{N}$ (resp. $t > 0$) and every $\lambda > 0$,

$$\begin{cases} \lambda \mathbb{P} \left(\sup_{0 \leq m \leq n} |X_m| > \lambda \right) \leq \mathbb{E}[X_0] + \mathbb{E}[X_n^-] & \text{For the discrete case} \\ \lambda \mathbb{P} \left(\sup_{0 \leq s \leq t} |X_s| > \lambda \right) \leq \mathbb{E}[X_0] + \mathbb{E}[X_t^-] & \text{For the continuous case} \end{cases}$$

3. (Doob's inequality in L^p for submartingales) Let $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) be

a nonnegative submartingale (with right-continuous sample paths in the continuous case). Then, for every $n \in \mathbb{N}$ (resp. $t > 0$) and every $p > 1$,

$$\begin{cases} \mathbb{E} \left[\sup_{0 \leq m \leq n} X_m^p \right] \leq \left(\frac{p}{p-1} \right)^p \mathbb{E}[|X_n|^p] & \text{For the discrete case} \\ \mathbb{E} \left[\sup_{0 \leq s \leq t} X_s^p \right] \leq \left(\frac{p}{p-1} \right)^p \mathbb{E}[|X_t|^p] & \text{For the continuous case} \end{cases}$$

4. (Doob's inequality in L^p for martingales) Let $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) be a martingale (with right-continuous sample paths in the continuous case). Then, for every $n \in \mathbb{N}$ (resp. $t > 0$) and every $p > 1$,

$$\begin{cases} \mathbb{E} \left[\sup_{0 \leq m \leq n} |X_m|^p \right] \leq \left(\frac{p}{p-1} \right)^p \mathbb{E}[|X_n|^p] & \text{For the discrete case} \\ \mathbb{E} \left[\sup_{0 \leq s \leq t} |X_s|^p \right] \leq \left(\frac{p}{p-1} \right)^p \mathbb{E}[|X_t|^p] & \text{For the continuous case} \end{cases}$$

Note that part (3) and (4) of the inequality is useful only if $\mathbb{E}[|X_n|^p] < \infty$ (resp. $\mathbb{E}[|X_t|^p] < \infty$).

Proof for Theorem.

- Let us prove the first assertion. Consider the stopping time

$$T = \inf\{k \geq 0 : X_k \geq a\}.$$

Then, if

$$A = \left\{ \sup_{0 \leq k \leq n} X_k \geq a \right\},$$

we have $A = \{T \leq n\}$. On the other hand, by applying the preceding lemma 1.2.12 to the stopping times $T \wedge n$ and n , we have

$$\mathbb{E}[X_{T \wedge n}] \leq \mathbb{E}[X_n].$$

Furthermore,

$$X_{T \wedge n} \geq a \mathbb{1}_A + X_n \mathbb{1}_{A^c}.$$

By combining these two inequalities, we get

$$\mathbb{E}[X_n] \geq a \mathbb{P}(A) + \mathbb{E}[X_n \mathbb{1}_{A^c}]$$

and it follows that $a \mathbb{P}(A) \leq \mathbb{E}[X_n \mathbb{1}_A]$, which gives the first assertion of the theorem in the discrete case.

For the continuous case, Fix $t > 0$ and consider a countable dense subset D of $\mathbb{R}_+$ such that $0 \in D$ and $t \in D$. Then $D \cap [0, t]$ is the increasing union of a sequence $(D_m)_{m \geq 1}$ of finite subsets $[0, t]$ of the form $D_m = \{t_0^m, t_1^m, \dots, t_m^m\}$ where

$0 = t_0^m < t_1^m < \dots < t_m^m = t$. For every fixed m , we can apply the discrete time case to the sequence $Y_n = X_{t_{n \wedge m}^m}$, which is a discrete submartingale with respect to the filtration $\mathcal{G}_n = \mathcal{F}_{t_{n \wedge m}^m}$. We get

$$\lambda \mathbb{P} \left(\sup_{s \in D_m} |X_s| > \lambda \right) \leq \mathbb{E}[X_t^+].$$

Then, we observe that

$$P \left(\sup_{s \in D_m} |X_s| > \lambda \right) \uparrow \mathbb{P} \left(\sup_{s \in D \cap [0, t]} |X_s| > \lambda \right)$$

when $m \uparrow \infty$. We have thus

$$\lambda \mathbb{P} \left(\sup_{s \in D \cap [0, t]} |X_s| > \lambda \right) \leq \mathbb{E}[X_t^+].$$

Finally, the right-continuity of sample paths (and the fact that $t \in D$) ensures that

$$\sup_{s \in D \cap [0, t]} |X_s| = \sup_{s \in [0, t]} |X_s|.$$

- The proof of the (2) is very similar to the proof of the (1). We begin with the discrete case and introduce the stopping time

$$R = \inf\{k \geq 0 : X_k \geq a\},$$

and the event $B = \{R \leq n\}$. We have then $\mathbb{E}[X_{R \wedge n}] \leq \mathbb{E}[X_0]$, and $\mathbb{E}[X_{R \wedge n}] \geq a\mathbb{P}(B) + \mathbb{E}[X_n \mathbf{1}_{B^c}]$. It follows that $a\mathbb{P}(B) \leq \mathbb{E}[X_0] - \mathbb{E}[X_n \mathbf{1}_{B^c}]$, which leads to the desired result. Then for the continuous case, we can use the same argument as in assertion (1) by considering a countable dense subset D of $\mathbb{R}_+$ such that $0 \in D$ and $t \in D$, and then applying the discrete case to the sequence $Y_n = X_{t_{n \wedge m}^m}$ for every fixed m .

- In order to prove the assertion (3) in the discrete case, we may assume that $\mathbb{E}[(X_n)^p] < \infty$. Then, by Jensen's inequality for conditional expectations, we have, for every $0 \leq k \leq n$,

$$\mathbb{E}[(X_k)^p] \leq \mathbb{E}[\mathbb{E}[X_n | \mathcal{F}_k]^p] \leq \mathbb{E}[\mathbb{E}[(X_n)^p | \mathcal{F}_k]] = \mathbb{E}[(X_n)^p].$$

Denote $\widetilde{X}_n = \sup_{0 \leq m \leq n} X_m$. Since $\widetilde{X}_n \leq X_0 + \dots + X_n$, it follows that we have also $\mathbb{E}[(\widetilde{X}_n)^p] < \infty$. By assertion (1) of the theorem, for every $a > 0$,

$$a \mathbb{P}(\widetilde{X}_n \geq a) \leq \mathbb{E}[X_n \mathbf{1}_{\{\widetilde{X}_n \geq a\}}].$$

We multiply each side of this inequality by a^{p-2} and integrate with respect to

Lebesgue measure on $(0, \infty)$. For the left side, we get

$$\int_0^\infty a^{p-1} \mathbb{P}(\widetilde{X}_n \geq a) da = \mathbb{E} \left[\int_0^{\widetilde{X}_n} a^{p-1} da \right] = \frac{1}{p} \mathbb{E}[(\widetilde{X}_n)^p]$$

using the Fubini theorem. Similarly, we get for the right side

$$\begin{aligned} \int_0^\infty a^{p-2} \mathbb{E}[X_n \mathbf{1}_{\{\widetilde{X}_n \geq a\}}] da &= \mathbb{E} \left[X_n \int_0^{\widetilde{X}_n} a^{p-2} da \right] \\ &= \frac{1}{p-1} \mathbb{E}[X_n (\widetilde{X}_n)^{p-1}] \\ &\leq \frac{1}{p-1} \mathbb{E}[(X_n)^p]^{\frac{1}{p}} \mathbb{E}[(\widetilde{X}_n)^p]^{\frac{p-1}{p}} \end{aligned}$$

by the Hölder inequality. We thus get

$$\frac{1}{p} \mathbb{E}[(\widetilde{X}_n)^p] \leq \frac{1}{p-1} \mathbb{E}[(X_n)^p]^{\frac{1}{p}} \mathbb{E}[(\widetilde{X}_n)^p]^{\frac{p-1}{p}}$$

giving the assertion (1) of the theorem (we use the fact that $\mathbb{E}[(\widetilde{X}_n)^p] < \infty$). Then for the continuous case, we can use the same argument as in assertion (1) by considering a countable dense subset D of $\mathbb{R}_+$ such that $0 \in D$ and $t \in D$, and then applying the discrete case to the sequence $Y_n = X_{t_n \wedge m}$ for every fixed m .

- The assertion (4) of the theorem follows from the assertion (3) applied to the submartingale $X_n = |Y_n|$.

■

In what follows, you will find that most of the results stated in the continuous time case requiring the right-continuity of the sample paths of the process, which is not that hard to verify in most of the cases thanks to the following theorem, which gives a sufficient condition to find a modification with the desired regularity of the sample paths.

Theorem 1.3.6: Sufficient condition for càdlàg modification

Assume that the filtration $(\mathcal{F}_t)$ is right-continuous and complete. Let $X = (X_t)_{t \geq 0}$ be a supermartingale, such that the function $t \mapsto \mathbb{E}[X_t]$ is right-continuous. Then X has a modification with càdlàg^a sample paths, which is also an $(\mathcal{F}_t)$ -supermartingale.

^a. refers to the french sentence "continue à droite et limite à gauche" which means "right-continuous and have left-limits"

Proof for Theorem.

■ The proof of this theorem could be found in [Stochastic] ■

Theorem 1.3.7: Doob's L^p convergence theorem

Let $(X_n)_{n \in \mathbb{N}}$ be a martingale and $p \in (1, \infty)$. Suppose that

$$\sup_{n \in \mathbb{N}} \mathbb{E}[|X_n|^p] < \infty,$$

Then, X_n converges a.s. and in L^p to a random variable X_∞ such that

$$\mathbb{E}[|X_\infty|^p] = \sup_{n \in \mathbb{N}} \mathbb{E}[|X_n|^p]$$

and we have

$$\mathbb{E}[(\sup_{n \in \mathbb{N}} |X_n|^p)] \leq \left(\frac{p}{p-1} \right)^p \mathbb{E}[|X_\infty|^p].$$

Proof for Theorem.

Since the martingale (X_n) is bounded in L^p , it is also bounded in L^1 , and we know from Theorem 1.3.9 that X_n converges a.s. to a limiting random variable X_∞ . Moreover, by setting $X_n^* = \sup_{1 \leq k \leq n} |X_k|$, the theorem 1.3.5 shows that, for every $n \in \mathbb{Z}_+$,

$$\mathbb{E}[(X_n^*)^p] \leq \left(\frac{p}{p-1} \right)^p \sup_{k \in \mathbb{Z}_+} \mathbb{E}[|X_k|^p].$$

Since $X_n^* \uparrow X_\infty^*$ when $n \uparrow \infty$, the monotone convergence theorem gives

$$\mathbb{E}[(X_\infty^*)^p] \leq \left(\frac{p}{p-1} \right)^p \sup_{k \in \mathbb{Z}_+} \mathbb{E}[|X_k|^p] < \infty$$

and thus $X_\infty^* \in L^p$. Since all random variables $|X_n|^p$ are dominated by $|X_\infty^*|^p$, the dominated convergence theorem ensures that X_n converges to X_∞ in L^p . Finally, we know from proof of (3) in theorem 1.3.5 that the sequence $\mathbb{E}[|X_n|^p]$ is increasing, and therefore

$$\mathbb{E}[|X_\infty|^p] = \lim_{n \rightarrow \infty} \mathbb{E}[|X_n|^p] = \sup_{n \in \mathbb{Z}_+} \mathbb{E}[|X_n|^p].$$

Theorem 1.3.8: Kolmogorov's criterion for a.s. convergence

Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of independent real random variables in L^2 . Assume that $\mathbb{E}[X_n] = 0$ for every $n \in \mathbb{N}$. Then the following two conditions are equivalent :

$$1. \sum_{n=0}^{\infty} \mathbb{E}[(X_n)^2] < \infty.$$

2. The series $\sum_{n=0}^{\infty} X_n$ converges a.s. and in L^2 .

Proof for Theorem.

For every integer $n \geq 0$, set $\mathcal{F}_n = \sigma(X_0, X_1, \dots, X_n)$ and

$$M_n = \sum_{k=0}^n X_k.$$

Since

$$\mathbb{E}[M_{n+1}|\mathcal{F}_n] = \sum_{k=0}^n X_k + \mathbb{E}[X_{n+1}] = \sum_{k=0}^n X_k,$$

$(M_n)_{n \in \mathbb{Z}_+}$ is a martingale with respect to the filtration $(\mathcal{F}_n)_{n \in \mathbb{Z}_+}$. Furthermore, since $\mathbb{E}[X_j X_k] = 0$ if $j \neq k$, one has

$$\mathbb{E}[(M_n)^2] = \sum_{k=0}^n \mathbb{E}[(X_k)^2].$$

Hence, if M_n converges in L^2 as $n \rightarrow \infty$, condition (1) of the theorem must hold (we have just recovered a special case of a classical result of Hilbert space theory, see Riesz Fischer Theorem ?? in the Appendix). Conversely, if (1) holds, the martingale M_n is bounded in L^2 , and by Theorem 1.3.7 we get that M_n converges a.s. and in L^2 . ■

1.3.3 Almost sure convergence

Theorem 1.3.9: Almost convergence of supermartingales

Let X be a supermartingale (with right-continuous sample paths if time is continuous). Assume that the collection $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) is bounded in L^1 . Then, it converge almost surely to a random variable $X_\infty \in L^1$.

Proof for Theorem.

We will prove it in the continuous time case, the discrete time case can be proven by similar arguments and it is left exercise for the reader. Let D be a countable dense subset of $\mathbb{R}_+$. Using lemma 1.3.4, we have, for every $a < b$ and every $T \in D$,

$$\mathbb{E} [M_{ab}^X(D \cap [0, T])] \leq \frac{1}{b-a} \mathbb{E} [(X_T - a)^-]$$

By Monotone convergence theorem, we get, for every $a < b$,

$$\mathbb{E} [M_{ab}^X(D)] \leq \frac{1}{b-a} \sup_{t \geq 0} \mathbb{E} [(X_t - a)^-] < +\infty.$$

Since the collection $(X_t)_{t \geq 0}$ is bounded in L^1 , the right-hand side of the previous display is finite. Hence, a.s. for all rationals $a < b$, we have $M_{ab}^X(D) < \infty$. It follows from

Lemma 1.3.3 that the limit

$$X_\infty := \lim_{D \ni t \rightarrow \infty} X_t$$

exists a.s. in $[-\infty, +\infty]$. We can in fact exclude the values $-\infty$ and $+\infty$ since, by Fatou's lemma,

$$\mathbb{E}[|X_\infty|] \leq \liminf_{D \ni t \rightarrow \infty} \mathbb{E}[|X_t|] < +\infty$$

and we get that $X_\infty \in L^1$. The right-continuity of X samples (which we have not used yet) allows us to remove the restriction to D in the limit. ■

Corollary 1.3.10

Let $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) be a nonnegative supermartingale. Then, X_n (resp. X_t) converges a.s. as $n \rightarrow \infty$ (resp. $t \rightarrow \infty$). Its limit X_∞ is in L^1 and we have $X_n \geq \mathbb{E}[X_\infty | \mathcal{F}_n]$ for every $n \in \mathbb{N}$. (resp. $X_t \geq \mathbb{E}[X_\infty | \mathcal{F}_t]$ for every $t \geq 0$).

Backward supermartingales A useful variant of the previous theorem is the following one, which is stated in a backward form in discrete time setup, we will not prove it however proof could be found in [Stochastic]. Since it will be used only in a proof step of supermartingale optional theorem in the continuous time case.

A backward filtration is a sequence $(\mathcal{F}_n)_{n \in \mathbb{Z}_-}$ of sub- σ -fields of $\mathcal{A}$, which is indexed by the set $\mathbb{Z}_- = \{0, -1, -2, \dots\}$ of all nonpositive integers, and is such that $\mathcal{F}_n \subset \mathcal{F}_m$ whenever $n, m \in \mathbb{Z}_-$ and $n \leq m$. We then set

$$\mathcal{F}_{-\infty} = \bigcap_{n \in \mathbb{Z}_-} \mathcal{F}_n$$

which is also a sub- σ -field of $\mathcal{A}$. It is important to observe that, in contrast with the "forward case" studied in the previous sections, the σ -field $\mathcal{F}_n$ becomes smaller and smaller when $n \downarrow -\infty$ (the larger $|n|$ is, the smaller the σ -field $\mathcal{F}_n$).

In what follows, we fix a backward filtration $(\mathcal{F}_n)_{n \in \mathbb{Z}_-}$. Let $X = (X_n)_{n \in \mathbb{Z}_-}$ be a sequence of real random variables indexed by $\mathbb{Z}_-$. We say that X is a backward martingale (resp. a backward supermartingale, a backward submartingale) if X_n is $\mathcal{F}_n$ -measurable and $\mathbb{E}[|X_n|] < \infty$ for every $n \in \mathbb{Z}_-$, and if, for every $m, n \in \mathbb{Z}_-$ such that $n \leq m$,

$$X_n = \mathbb{E}[X_m | \mathcal{F}_n] \quad (\text{resp. } X_n \geq \mathbb{E}[X_m | \mathcal{F}_n], X_n \leq \mathbb{E}[X_m | \mathcal{F}_n]).$$

Theorem 1.3.11: Backward supermartingale convergence

Let $(X_n)_{n \in \mathbb{Z}_-}$ be a backward supermartingale. Suppose that

$$\sup_{n \in \mathbb{Z}_-} \mathbb{E}[|X_n|] < \infty.$$

Then the sequence $(X_n)_{n \in \mathbb{Z}_-}$ is uniformly integrable and converges a.s. and in L^1 to

a random variable X_∞ as $n \rightarrow -\infty$. Moreover for every $n \in \mathbb{Z}_-$,

$$\mathbb{E}[X_n | \mathcal{F}_{-\infty}] \leq X_\infty,$$

and equality holds in the last display if $(X_n)_{n \in \mathbb{Z}_-}$ is a backward martingale.

1.3.4 Martingale Closure

Definition 1.3.12: Closed Martingale

A martingale $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) is said to be *closed* if there exists a random variable $Z \in L^1$ such that, for every $n \in \mathbb{N}$ (resp. $t \geq 0$),

$$\begin{cases} X_n = \mathbb{E}[Z | \mathcal{F}_n] & \text{For the discrete time} \\ X_t = \mathbb{E}[Z | \mathcal{F}_t] & \text{For the continuous time} \end{cases}$$

Definition 1.3.13: Uniform integrability

A collection $(X_i)_{i \in I}$ of random variables in $L^1(\Omega, \mathcal{A}, \mathbb{P})$ is said to be *uniformly integrable* (u.i. in short) if

$$\lim_{a \rightarrow +\infty} \left(\sup_{i \in I} \mathbb{E}[|X_i| \mathbf{1}_{\{|X_i| > a\}}] \right) = 0.$$

A uniformly integrable collection $(X_i)_{i \in I}$ is bounded in L^1 . To verify this, take a large enough so that

$$\sup_{i \in I} \mathbb{E}[|X_i| \mathbf{1}_{\{|X_i| > a\}}] \leq 1$$

and write $\mathbb{E}[|X_i|] \leq \mathbb{E}[|X_i| \mathbf{1}_{\{|X_i| \leq a\}}] + \mathbb{E}[|X_i| \mathbf{1}_{\{|X_i| > a\}}] \leq a + 1$. The converse is false : a collection $(X_i)_{i \in I}$ which is bounded in L^1 may not be uniformly integrable.

Proposition 1.3.14

Let $(X_i)_{i \in I}$ be a bounded subset of L^1 . The following two conditions are equivalent :

1. The collection $(X_i)_{i \in I}$ is u.i.
2. For every $\varepsilon > 0$, one can find $\delta > 0$ such that, for every event $A \in \mathcal{A}$ of probability $\mathbb{P}(A) < \delta$, one has

$$\forall i \in I, \quad \mathbb{E}[|X_i| \mathbf{1}_A] < \varepsilon.$$

Proof for Proposition.

(1) $\Rightarrow$ (2) Let $\varepsilon > 0$. We can first fix $a > 0$ such that

$$\sup_{i \in I} \mathbb{E}[|X_i| \mathbf{1}_{\{|X_i| > a\}}] < \frac{\varepsilon}{2}.$$

If we set $\delta = \varepsilon/(2a)$, then the condition $\mathbb{P}(A) < \delta$ implies, that, for every $i \in I$,

$$\mathbb{E}[|X_i| \mathbf{1}_A] \leq \mathbb{E}[|X_i| \mathbf{1}_{A \cap \{|X_i| \leq a\}}] + \mathbb{E}[|X_i| \mathbf{1}_{\{|X_i| > a\}}] \leq a\mathbb{P}(A) + \frac{\varepsilon}{2} < \varepsilon.$$

(2) $\Rightarrow$ (1) Set $C = \sup_{i \in I} \mathbb{E}[|X_i|]$. By the Markov inequality, we have for every $a > 0$,

$$\forall i \in I, \quad \mathbb{P}(|X_i| > a) \leq \frac{C}{a}.$$

Let $\varepsilon > 0$ and choose $\delta > 0$ so that the property stated in (2) holds. Then, if a is large enough so that $C/a < \delta$, we have $\mathbb{P}(|X_i| > a) < \delta$ for every $i \in I$, and thus, by (2),

$$\forall i \in I, \quad \mathbb{E}[|X_i| \mathbf{1}_{\{|X_i| > a\}}] < \varepsilon.$$

We conclude that the collection $(X_i)_{i \in I}$ is u.i. ■

Theorem 1.3.15: Vitali convergence theorem

Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of random variables in L^1 that converges to X_∞ in probability. The following two conditions are equivalent :

1. The sequence $(X_n)_{n \in \mathbb{N}}$ converges to X_∞ in L^1 .
2. The sequence $(X_n)_{n \in \mathbb{N}}$ is uniformly integrable.

Proof for Theorem.

- (1) $\Rightarrow$ (2) If the sequence $(X_n)_{n \in \mathbb{N}}$ converges in L^1 , then it is bounded in L^1 . Then, let $\varepsilon > 0$. We can choose N large enough so that, for every $n \geq N$,

$$\mathbb{E}[|X_n - X_N|] < \frac{\varepsilon}{2}.$$

Since the finite set $\{X_1, \dots, X_N\}$ is u.i., Proposition 1.3.14 allows us to choose $\delta > 0$ such that, for every event A of probability $\mathbb{P}(A) < \delta$, we have

$$\forall n \in \{1, \dots, N\}, \quad \mathbb{E}[|X_n| \mathbf{1}_A] < \frac{\varepsilon}{2}.$$

Then, if $n > N$, we have also

$$\mathbb{E}[|X_n| \mathbf{1}_A] \leq \mathbb{E}[|X_N| \mathbf{1}_A] + \mathbb{E}[|X_n - X_N|] < \varepsilon.$$

By Proposition 1.3.14, the sequence $(X_n)_{n \in \mathbb{N}}$ is uniformly integrable.

- (2) $\Rightarrow$ (1) Using the characterization of uniform integrability in Proposition 1.3.14(ii), it is straightforward to verify that the collection $(X_n - X_m)_{n, m \in \mathbb{N}}$

is also u.i. Hence, if $\varepsilon > 0$ is fixed, we can choose $a > 0$ large enough so that, for every $m, n \in \mathbb{N}$,

$$\mathbb{E}[|X_n - X_m| \mathbf{1}_{\{|X_n - X_m| > a\}}] < \varepsilon.$$

Then, for every $m, n \in \mathbb{N}$,

$$\begin{aligned} \mathbb{E}[|X_n - X_m|] &\leq \mathbb{E}[|X_n - X_m| \mathbf{1}_{\{|X_n - X_m| \leq \varepsilon\}}] + \mathbb{E}[|X_n - X_m| \mathbf{1}_{\{\varepsilon < |X_n - X_m| \leq a\}}] \\ &\quad + \mathbb{E}[|X_n - X_m| \mathbf{1}_{\{|X_n - X_m| > a\}}] \\ &\leq 2\varepsilon + a \mathbb{P}(|X_n - X_m| > \varepsilon). \end{aligned}$$

The convergence in probability of the sequence $(X_n)_{n \in \mathbb{N}}$ implies that

$$\mathbb{P}(|X_n - X_m| > \varepsilon) \leq \mathbb{P}\left(|X_n - X_\infty| > \frac{\varepsilon}{2}\right) + \mathbb{P}\left(|X_m - X_\infty| > \frac{\varepsilon}{2}\right) \xrightarrow{n, m \rightarrow \infty} 0.$$

It follows that

$$\limsup_{m, n \rightarrow \infty} \mathbb{E}[|X_n - X_m|] \leq 2\varepsilon$$

and, since ε was arbitrary, this shows that the sequence $(X_n)_{n \in \mathbb{N}}$ is Cauchy in L^1 , hence converges in L^1 . ■

Theorem 1.3.16: Doob's Uniform Integrability convergence theorem

Let X be a martingale (with right-continuous sample paths if time is continuous). Then the following properties are equivalent :

1. X is closed ;
2. the collection $(X_n)_{n \in \mathbb{N}}$ ($(X_t)_{t \geq 0}$ if time is continuous) is uniformly integrable ;
3. X_n (resp. X_t) converges a.s. and in L^1 as $n \rightarrow \infty$ (resp. $t \rightarrow \infty$).

Moreover, if these properties hold, we have $X_n = \mathbb{E}[X_\infty | \mathcal{F}_n]$ (resp. $X_t = \mathbb{E}[X_\infty | \mathcal{F}_t]$) for every $n \in \mathbb{N}$ (resp. $t \geq 0$), where $X_\infty \in L^1$ is the a.s. limit of X_n (resp. X_t) as $n \rightarrow \infty$ (resp. $t \rightarrow \infty$).

Proof for Theorem.

- (1) $\Rightarrow$ (2) : if $Z \in L^1$, the collection of all random variable $\mathbb{E}[Z | \mathcal{G}]$ when $\mathcal{G}$ varies among all sub- σ -fields of $\mathcal{F}_\infty$ is u.i.
- (2) $\Rightarrow$ (3) : If (2) holds, then $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) is bounded in L^1 , and thus converges a.s. to some random variable X_∞ by Theorem 1.3.9. By uniform integrability, the latter convergence also holds in L^1 .
- (3) $\Rightarrow$ (1) : If (3) holds, for every $m \in \mathbb{N}$ (resp. $s \geq 0$), we can pass to the limit $n \rightarrow +\infty$ (resp. $t \rightarrow +\infty$) in the equality, $X_m = \mathbb{E}(X_n | \mathcal{F}_m)$ (resp. $X_s = \mathbb{E}(X_t | \mathcal{F}_s)$) and using the theorem ?? we get $X_m = \mathbb{E}(X_\infty | \mathcal{F}_m)$ (resp. $X_s = \mathbb{E}(X_\infty | \mathcal{F}_s)$), which gives the desired result.

Corollary 1.3.17

Let $Z \in L^1(\Omega, \mathcal{A}, \mathbb{P})$. The martingale $X_n = \mathbb{E}[Z \mid \mathcal{F}_n]$ (resp. $X_t = \mathbb{E}[Z \mid \mathcal{F}_t]$) converges a.s. and in L^1 to $X_\infty = \mathbb{E}[Z \mid \mathcal{F}_\infty]$.

1.4 Optional stopping theorems**1.4.1 The theorem for martingales****Theorem 1.4.1: Optional stopping theorem – martingale**

Let $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) be a uniformly integrable martingale (with right-continuous sample paths if time is continuous). Let S and T be two stopping times with $S \leq T$. Then X_S and X_T are in L^1 and

$$X_S = \mathbb{E}[X_T \mid \mathcal{F}_S].$$

Proof for Theorem.

We begin with the discrete time case. From the preceding section, we know that the martingale $(X_n)_{n \in \mathbb{N}}$ is closed and $X_n = \mathbb{E}[X_\infty \mid \mathcal{F}_n]$ for every n . Let us verify that $X_T \in L^1$:

$$\begin{aligned} \mathbb{E}[|X_T|] &= \sum_{n=0}^{\infty} \mathbb{E}[\mathbf{1}_{\{T=n\}} |X_n|] + \mathbb{E}[\mathbf{1}_{\{T=\infty\}} |X_\infty|] \\ &= \sum_{n=0}^{\infty} \mathbb{E}[\mathbf{1}_{\{T=n\}} |\mathbb{E}[X_\infty \mid \mathcal{F}_n]|] + \mathbb{E}[\mathbf{1}_{\{T=\infty\}} |X_\infty|] \\ &\leq \sum_{n=0}^{\infty} \mathbb{E}[\mathbf{1}_{\{T=n\}} \mathbb{E}[|X_\infty| \mid \mathcal{F}_n]] + \mathbb{E}[\mathbf{1}_{\{T=\infty\}} |X_\infty|] \\ &= \sum_{n=0}^{\infty} \mathbb{E}[\mathbf{1}_{\{T=n\}} |X_\infty|] + \mathbb{E}[\mathbf{1}_{\{T=\infty\}} |X_\infty|] \\ &= \mathbb{E}[|X_\infty|] < \infty. \end{aligned}$$

Then, if $A \in \mathcal{F}_T$,

$$\begin{aligned}
\mathbb{E}[\mathbf{1}_A X_T] &= \sum_{n \in \mathbb{N} \cup \{\infty\}} \mathbb{E}[\mathbf{1}_{A \cap \{T=n\}} X_T] \\
&= \sum_{n \in \mathbb{N} \cup \{\infty\}} \mathbb{E}[\mathbf{1}_{A \cap \{T=n\}} X_n] \\
&= \sum_{n \in \mathbb{N} \cup \{\infty\}} \mathbb{E}[\mathbf{1}_{A \cap \{T=n\}} X_\infty] \\
&= \mathbb{E}[\mathbf{1}_A X_\infty].
\end{aligned}$$

In the first equality, we use the fact that $X_T \in L^1$ to justify the interchange of sum and expectation via the Fubini theorem. In the third equality, we use the properties $X_n = \mathbb{E}[X_\infty | \mathcal{F}_n]$ and $A \cap \{T = n\} \in \mathcal{F}_n$ for $A \in \mathcal{F}_T$. Since X_T is $\mathcal{F}_T$ -measurable and in L^1 , the preceding display shows that X_T satisfies the characteristic property of the conditional expectation $\mathbb{E}[X_\infty | \mathcal{F}_T]$.

Finally, if S and T are two stopping times such that $S \leq T$, the property $\mathcal{F}_S \subset \mathcal{F}_T$ (Proposition 1.2.5) and Theorem of nested σ -fields in conditional expectation ?? imply that

$$X_S = \mathbb{E}[X_\infty | \mathcal{F}_S] = \mathbb{E}[\mathbb{E}[X_\infty | \mathcal{F}_T] | \mathcal{F}_S] = \mathbb{E}[X_T | \mathcal{F}_S].$$

For the continuous time case, Set, for every integer $n \geq 0$,

$$T_n = \frac{\lceil 2^n T \rceil}{2^n} \cdot \mathbf{1}_{\{T < \infty\}} + \infty \cdot \mathbf{1}_{\{T = \infty\}}, \quad n = 0, 1, 2, \dots$$

and similarly

$$S_n = \frac{\lceil 2^n S \rceil}{2^n} \cdot \mathbf{1}_{\{S < \infty\}} + \infty \cdot \mathbf{1}_{\{S = \infty\}}, \quad n = 0, 1, 2, \dots$$

By Proposition 1.2.9, (T_n) and (S_n) are two sequences of stopping times that decrease respectively to T and to S . Moreover, we have $S_n \leq T_n$ for every $n \geq 0$.

Now observe that, for every fixed n , $2^n S_n$ and $2^n T_n$ are stopping times of the discrete filtration $\mathcal{H}_k^{(n)} := \mathcal{F}_{k/2^n}$, and $Y_k^{(n)} := X_{k/2^n}$ is a discrete martingale with respect to this filtration. From the discrete case we proven before, we get that $Y_{2^n S_n}^{(n)}$ and $Y_{2^n T_n}^{(n)}$ are in L^1 , and

$$X_{S_n} = Y_{2^n S_n}^{(n)} = \mathbb{E}[Y_{2^n T_n}^{(n)} | \mathcal{H}_{2^n S_n}^{(n)}] = \mathbb{E}[X_{T_n} | \mathcal{F}_{S_n}]$$

(here we need to verify that $\mathcal{H}_{2^n S_n}^{(n)} = \mathcal{F}_{S_n}$, but this is straightforward).

Let $A \in \mathcal{F}_S$. Since $\mathcal{F}_S \subset \mathcal{F}_{S_n}$, we have $A \in \mathcal{F}_{S_n}$ and thus

$$\mathbb{E}[\mathbf{1}_A X_{S_n}] = \mathbb{E}[\mathbf{1}_A X_{T_n}].$$

By the right-continuity of sample paths, we get a.s.

$$X_S = \lim_{n \rightarrow \infty} X_{S_n}, \quad X_T = \lim_{n \rightarrow \infty} X_{T_n}.$$

These limits also hold in L^1 . Indeed, thanks again to the optional stopping theorem for

uniformly integrable discrete martingales, we have $X_{S_n} = \mathbb{E}[X_\infty | \mathcal{F}_{S_n}]$ for every n , and thus the sequence (X_{S_n}) is uniformly integrable (and the same holds for the sequence (X_{T_n})).

The L^1 -convergence implies that X_S and X_T belong to L^1 , and also allows us to pass to the limit $n \rightarrow \infty$ in the equality $\mathbb{E}[\mathbf{1}_A X_{S_n}] = \mathbb{E}[\mathbf{1}_A X_{T_n}]$ in order to get

$$\mathbb{E}[\mathbf{1}_A X_S] = \mathbb{E}[\mathbf{1}_A X_T].$$

Since this holds for every $A \in \mathcal{F}_S$, and since the variable X_S is $\mathcal{F}_S$ -measurable (by the remarks before the theorem), we conclude that

$$X_S = \mathbb{E}[X_T | \mathcal{F}_S],$$

which completes the proof. ■

In particular, for every stopping time S , we have

$$X_S = \mathbb{E}[X_\infty | \mathcal{F}_S],$$

and

$$\mathbb{E}[X_S] = \mathbb{E}[X_\infty] = \mathbb{E}[X_0].$$

Corollary 1.4.2

If T and S are stopping time such that $S \leq T$, and both S and T are bounded, then X_S and X_T are in L^1 and

$$X_S = \mathbb{E}[X_T | \mathcal{F}_S].$$

Proof for Corollary.

Let $a \geq 0$ such that $S \leq T \leq a$. We apply Optional Stopping Theorem 1.4.1 to the martingale $\{X_{n \wedge a}\}_{n \geq 0}$ (resp. $\{X_{t \wedge a}\}_{t \geq 0}$) which is closed by X_a . ■

Corollary 1.4.3

Let $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) be a martingale (with right-continuous sample paths if time is continuous), and let T be a stopping time.

1. The process $(X_{n \wedge T})_{n \in \mathbb{N}}$ (resp. $(X_{t \wedge T})_{t \geq 0}$) is still a martingale.
2. Suppose in addition that the martingale $(X_n)_{n \in \mathbb{N}}$ (resp. $(X_t)_{t \geq 0}$) is uniformly integrable. Then the process $(X_{n \wedge T})_{n \in \mathbb{N}}$ (resp. $(X_{t \wedge T})_{t \geq 0}$) is also a uniformly integrable martingale, and more precisely we have,

$$\begin{cases} X_{n \wedge T} = \mathbb{E}[X_T | \mathcal{F}_{n \wedge T}], & n \in \mathbb{N} \quad \text{for the discrete case,} \\ X_{t \wedge T} = \mathbb{E}[X_T | \mathcal{F}_{t \wedge T}], & t \geq 0 \quad \text{for the continuous case.} \end{cases}$$

Proof for Corollary.

We will prove it for the continuous time case, the discrete time case can be proven by similar arguments and it is left exercise for the reader. We start with the proof of (2). Note that $t \wedge T$ is a stopping time by property (f) of stopping times. By Theorem 1.4.1, $X_{t \wedge T}$ and X_T are in L^1 , and we also know that $X_{t \wedge T}$ is $\mathcal{F}_{t \wedge T}$ -measurable, hence $\mathcal{F}_t$ -measurable since $\mathcal{F}_{t \wedge T} \subset \mathcal{F}_t$. So in order to get formula in (2), it is enough to prove that, for every $A \in \mathcal{F}_t$,

$$\mathbb{E}[\mathbb{1}_A X_T] = \mathbb{E}[\mathbb{1}_A X_{t \wedge T}].$$

Let us fix $A \in \mathcal{F}_t$. First, we have trivially

$$\mathbb{E}[\mathbb{1}_{A \cap \{T \leq t\}} X_T] = \mathbb{E}[\mathbb{1}_{A \cap \{T \leq t\}} X_{t \wedge T}]. \quad (*)$$

On the other hand, by Theorem 1.4.1, we have

$$X_{t \wedge T} = \mathbb{E}[X_T \mid \mathcal{F}_{t \wedge T}],$$

and we notice that we have both $A \cap \{T > t\} \in \mathcal{F}_t$ and $A \cap \{T > t\} \in \mathcal{F}_T$ (the latter as a straightforward consequence of the definition of $\mathcal{F}_T$), so that $A \cap \{T > t\} \in \mathcal{F}_t \cap \mathcal{F}_T = \mathcal{F}_{t \wedge T}$. Using the preceding display, we obtain

$$\mathbb{E}[\mathbb{1}_{A \cap \{T > t\}} X_T] = \mathbb{E}[\mathbb{1}_{A \cap \{T > t\}} X_{t \wedge T}].$$

By adding this equality to (*), we get the desired result.

To prove (1), we just need to apply (2) to the (uniformly integrable) martingale $(X_{t \wedge a})_{a \geq 0}$, for any choice of $a \geq 0$. ■

1.4.2 The theorem for supermartingales**Theorem 1.4.4: Optional stopping theorem - supermartingale**

Let $(Z_n)_{n \in \mathbb{N}}$ (resp. $(Z_t)_{t \geq 0}$) be a nonnegative supermartingale (with right-continuous sample paths if time is continuous). Let U and V be two stopping times such that $U \leq V$. Then Z_U and Z_V are in L^1 , and

$$Z_U \geq \mathbb{E}[Z_V \mid \mathcal{F}_U].$$

Proof for Theorem.

Let's begin with the discrete time case. We first note that $(X_n)_{n \in \mathbb{N}}$ is bounded in L^1 . Theorem 1.3.9 then shows that X_n converges a.s. to X_∞ , so that the definition of X_T makes sense for any stopping time, even on the event $\{T = \infty\}$. If T is a bounded stopping time, we know that $\mathbb{E}[X_T] \leq \mathbb{E}[X_0]$ (Lemma 1.2.11). Fatou's lemma then

shows that, for any stopping time T ,

$$\mathbb{E}[X_T] \leq \liminf_{k \rightarrow \infty} \mathbb{E}[X_{T \wedge k}] \leq \mathbb{E}[X_0]$$

and thus $X_T \in L^1$.

Then let S and T be two stopping times such that $S \leq T$. Let us first assume that $T \leq N$ for some fixed integer N . According to Lemma 1.2.12, we have then $\mathbb{E}[X_S] \geq \mathbb{E}[X_T]$. More generally, let $A \in \mathcal{F}_S$, and set

$$S^A(\omega) = \begin{cases} S(\omega) & \text{if } \omega \in A, \\ N & \text{if } \omega \notin A. \end{cases}$$

It is straightforward to verify that S^A is a stopping time (just note that $\{S^A \leq n\} = \Omega$ if $n \geq N$, and $\{S^A \leq n\} = A \cap \{S \leq n\} \in \mathcal{F}_n$ if $n < N$, using the definition of $\mathcal{F}_S$). We define T^A in a similar manner, and note that T^A is also a stopping time since $A \in \mathcal{F}_S \subset \mathcal{F}_T$. Since $S^A \leq T^A$, we have $\mathbb{E}[X_{S^A}] \geq \mathbb{E}[X_{T^A}]$, and it follows that

$$\mathbb{E}[X_S \mathbf{1}_A] \geq \mathbb{E}[X_T \mathbf{1}_A].$$

Let us come back to the general case where S and T satisfy $S \leq T$ but are no longer assumed to be bounded. Let $A \in \mathcal{F}_S$, and $k \in \mathbb{N}$. Then $A \cap \{S \leq k\}$ belongs both to $\mathcal{F}_k$ (by the definition of $\mathcal{F}_S$) and to $\mathcal{F}_S$ (recall that S is $\mathcal{F}_S$ -measurable). From Proposition 1.2.5, $A \cap \{S \leq k\}$ belongs to $\mathcal{F}_{S \wedge k}$. Hence, by applying the bounded case to the stopping times $S \wedge k$ and $T \wedge k$, we get

$$\mathbb{E}[X_S \mathbf{1}_{A \cap \{S \leq k\}}] = \mathbb{E}[X_{S \wedge k} \mathbf{1}_{A \cap \{S \leq k\}}] \geq \mathbb{E}[X_{T \wedge k} \mathbf{1}_{A \cap \{S \leq k\}}].$$

By monotone convergence,

$$\lim_{k \rightarrow \infty} \uparrow \mathbb{E}[X_S \mathbf{1}_{A \cap \{S \leq k\}}] = \mathbb{E}[X_S \mathbf{1}_{A \cap \{S < \infty\}}].$$

On the other hand, Fatou's lemma gives

$$\liminf_{k \rightarrow \infty} \mathbb{E}[X_{T \wedge k} \mathbf{1}_{A \cap \{S \leq k\}}] \geq \mathbb{E}[X_T \mathbf{1}_{A \cap \{S < \infty\}}].$$

So we have obtained that

$$\mathbb{E}[X_S \mathbf{1}_{A \cap \{S < \infty\}}] \geq \mathbb{E}[X_T \mathbf{1}_{A \cap \{S < \infty\}}].$$

Since $X_S = X_T = X_\infty$ on the event $\{S = \infty\}$, it follows that

$$\mathbb{E}[X_S \mathbf{1}_A] \geq \mathbb{E}[X_T \mathbf{1}_A] = \mathbb{E}[\mathbb{E}[X_T \mid \mathcal{F}_S] \mathbf{1}_A].$$

Since this holds for every $A \in \mathcal{F}_S$, and both X_S and $\mathbb{E}[X_T \mid \mathcal{F}_S]$ are $\mathcal{F}_S$ -measurable, it follows that $X_S \geq \mathbb{E}[X_T \mid \mathcal{F}_S]$ a.s. (apply the last display with the set $A = \{X_S < \mathbb{E}[X_T \mid \mathcal{F}_S]\}$). This completes the proof of the theorem in the discrete case.

Now the continuous time case, Firstly, we make the extra assumption that U and V

are bounded and we verify that we then have $\mathbb{E}[Z_U] \geq \mathbb{E}[Z_V]$. Let $p \geq 1$ be an integer such that $U \leq p$ and $V \leq p$. For every integer $n \geq 0$, set

$$U_n = \sum_{k=0}^{p2^n-1} \frac{k+1}{2^n} \mathbf{1}_{\{k2^{-n} < U \leq (k+1)2^{-n}\}}, \quad V_n = \sum_{k=0}^{p2^n-1} \frac{k+1}{2^n} \mathbf{1}_{\{k2^{-n} < V \leq (k+1)2^{-n}\}}$$

in such a way (by Proposition 1.2.9) that (U_n) and (V_n) are two sequences of bounded stopping times that decrease respectively to U and V , and additionally we have $U_n \leq V_n$ for every $n \geq 0$. The right-continuity of sample paths ensures that $Z_{U_n} \rightarrow Z_U$ and $Z_{V_n} \rightarrow Z_V$ a.s. as $n \rightarrow \infty$. Then, by the optional stopping theorem for discrete supermartingales in the case of bounded stopping times (which we just proved), with respect to the filtration $(\mathcal{F}_{k/2^{n+1}})_{k \geq 0}$, we have for every $n \geq 0$,

$$Z_{U_{n+1}} \geq \mathbb{E}[Z_{U_n} \mid \mathcal{F}_{U_{n+1}}].$$

Setting $Y_n = Z_{U_{-n}}$ and $\mathcal{H}_n = \mathcal{F}_{U_{-n}}$, for every integer $n \leq 0$, we get that the sequence $(Y_n)_{n \leq 0}$ is a backward supermartingale with respect to the filtration $(\mathcal{H}_n)_{n \leq 0}$. Since, for every $n \geq 0$, $\mathbb{E}[Z_{U_n}] \leq \mathbb{E}[Z_0]$ (by another application of the discrete optional stopping theorem), the sequence $(Y_n)_{n \leq 0}$ is bounded in L^1 , and by the convergence theorem for backward supermartingales in Theorem 1.3.11, it converges in L^1 . Hence the convergence of Z_{U_n} to Z_U also holds in L^1 and similarly the convergence of Z_{V_n} to Z_V holds in L^1 . Since $U_n \leq V_n$, by yet another application of the discrete optional stopping theorem, we have $\mathbb{E}[Z_{U_n}] \geq \mathbb{E}[Z_{V_n}]$. Using the L^1 -convergence of Z_{U_n} and Z_{V_n} we can pass to the limit $n \rightarrow \infty$ and obtain that $\mathbb{E}[Z_U] \geq \mathbb{E}[Z_V]$ as claimed.

Let us prove the statement of the theorem (no longer assuming that U and V are bounded). By the first step of the proof applied to the stopping times 0 and $U \wedge p$, we have $\mathbb{E}[Z_{U \wedge p}] \leq \mathbb{E}[Z_0]$ for every $p \geq 1$, and Fatou's lemma gives $\mathbb{E}[Z_U] \leq \mathbb{E}[Z_0] < \infty$ and similarly $\mathbb{E}[Z_V] < \infty$. Fix $A \in \mathcal{F}_U \subset \mathcal{F}_V$ and recall our notation U^A for the stopping time defined by $U^A(\omega) = U(\omega)$ if $\omega \in A$ and $U^A(\omega) = \infty$ otherwise (cf. property (d) of stopping times). By the first part of the proof, we have, for every $p \geq 1$,

$$\mathbb{E}[Z_{U^A \wedge p}] \geq \mathbb{E}[Z_{V^A \wedge p}].$$

By writing each of these two expectations as a sum of expectations over the sets A^c , $A \cap \{U \leq p\}$ and $A \cap \{U > p\}$, and noting that $U > p$ implies $V > p$, we get

$$\mathbb{E}[Z_U \mathbf{1}_{A \cap \{U \leq p\}}] \geq \mathbb{E}[Z_V \mathbf{1}_{A \cap \{U \leq p\}}].$$

Letting $p \rightarrow \infty$ then gives

$$\mathbb{E}[Z_U \mathbf{1}_{A \cap \{U < \infty\}}] \geq \mathbb{E}[Z_V \mathbf{1}_{A \cap \{U < \infty\}}].$$

is trivial and by adding it to the preceding display, we obtain

$$\mathbb{E}[Z_U \mathbf{1}_A] \geq \mathbb{E}[Z_V \mathbf{1}_A] = \mathbb{E}[\mathbb{E}[Z_V \mid \mathcal{F}_U] \mathbf{1}_A].$$

Since this holds for every $A \in \mathcal{F}_U$ and Z_U is $\mathcal{F}_U$ -measurable, the desired result follows. ■

Remark This implies that $\mathbb{E}[Z_U] \geq \mathbb{E}[Z_V]$, and since $Z_U = Z_V = Z_\infty$ on the event $\{U = \infty\}$, it also follows that

$$\mathbb{E}[\mathbf{1}_{\{U < \infty\}} Z_U] \geq \mathbb{E}[\mathbf{1}_{\{U < \infty\}} Z_V] \geq \mathbb{E}[\mathbf{1}_{\{V < \infty\}} Z_V].$$

1.5 Finite Variation Processes

1.5.1 Finite Variation Functions

In our discussion of functions with finite variation, we restrict our attention to *continuous* functions, as this is the case of interest in the subsequent developments. Recall that a **signed measure** on a compact interval $[0, T]$ is the difference of two finite positive measures on $[0, T]$.

Definition 1.5.1: Finite Variation function

Let $T \geq 0$. A continuous function $a : [0, T] \rightarrow \mathbb{R}$ such that $a(0) = 0$ is said to have *finite variation* if there exists a signed measure μ on $[0, T]$ such that $a(t) = \mu([0, t])$ for every $t \in [0, T]$.

The measure μ is then determined uniquely by a . Since a is continuous and $a(0) = 0$, it follows that μ has no atoms¹.

By the Jordan Decomposition Theorem, any signed measure can be split uniquely as $\mu = \mu^+ - \mu^-$, where μ^+ and μ^- are mutually singular positive measures, ensuring that the structural decomposition of such restricted systems is completely rigid and unique. The decomposition of a as a difference of two finite positive measures on $[0, T]$ is not unique, but there exists a unique decomposition $\mu = \mu^+ - \mu^-$ such that μ^+ and μ^- are supported on disjoint Borel sets (D_+ and D_-). We write $|\mu|$ for the (finite) positive measure $|\mu| = \mu^+ + \mu^-$. The measure $|\mu|$ is called the *total variation* of a . We have $|\mu(A)| \leq |\mu|(A)$ for every $A \in \mathcal{B}([0, T])$. Moreover, the Radon-Nikodym derivative of μ with respect to $|\mu|$ is

$$\frac{d\mu}{d|\mu|} = \mathbf{1}_{D_+} - \mathbf{1}_{D_-}.$$

The fact that $a(t) = \mu_+([0, t]) - \mu_-([0, t])$ shows that a is the difference of two monotone nondecreasing continuous functions that vanish at 0 (since μ has no atoms, the same holds for μ_+ or μ_-). Conversely, the difference of two monotone nondecreasing continuous functions that vanish at 0 has finite variation in the sense of the previous definition. Indeed, this follows from the well-known fact that the formula $g(t) = \theta([0, t])$, $t \in [0, T]$ induces a bijection between monotone nondecreasing right-continuous functions $g : [0, T] \rightarrow \mathbb{R}_+$ and finite positive measures θ on $[0, T]$.

1. A measure μ has an atom A if $\mu(A) > 0$ and for every $B \in \mathcal{A}$ with $B \subseteq A$, either $\mu(B) = 0$ or $\mu(B) = \mu(A)$.

Let $f : [0, T] \rightarrow \mathbb{R}$ be a measurable function such that $\int_{[0, T]} |f(s)| |\mu|(ds) < \infty$. We set

$$\begin{aligned} \int_0^T f(s) da(s) &= \int_{[0, T]} f(s) \mu(ds), \\ \int_0^T f(s) |da(s)| &= \int_{[0, T]} f(s) |\mu|(ds). \end{aligned}$$

Then the bound

$$\left| \int_0^T f(s) da(s) \right| \leq \int_0^T |f(s)| |da(s)|$$

holds. By restricting a to $[0, t]$ (which amounts to restricting μ, μ_+, μ_-), we can define $\int_0^t f(s) da(s)$ for every $t \in [0, T]$, and we observe that the function $t \mapsto \int_0^t f(s) da(s)$ also has finite variation on $[0, T]$ (the associated measure is just $\mu'(ds) = f(s)\mu(ds)$).

Proposition 1.5.2

For every $t \in (0, T]$,

$$\int_0^t |da(s)| = \sup \left\{ \sum_{i=1}^p |a(t_i) - a(t_{i-1})| \right\},$$

where the supremum is over all subdivisions $0 = t_0 < t_1 < \dots < t_p = t$ of $[0, t]$. More precisely, for any increasing sequence $0 = t_0^n < t_1^n < \dots < t_{p_n}^n = t$ of subdivisions of $[0, t]$ whose mesh tends to 0, we have

$$\lim_{n \rightarrow \infty} \sum_{i=1}^{p_n} |a(t_i^n) - a(t_{i-1}^n)| = \int_0^t |da(s)|.$$

Proof for Proposition.

Clearly, it is enough to treat the case $t = T$. The inequality $\geq$ in the first assertion is very easy since, for any subdivision $0 = t_0 < t_1 < \dots < t_p = T$ of $[0, T]$,

$$|a(t_i) - a(t_{i-1})| = |\mu((t_{i-1}, t_i])| \leq |\mu|((t_{i-1}, t_i]), \quad \forall i \in \{1, \dots, p\},$$

and

$$\sum_{i=1}^p |\mu|((t_{i-1}, t_i]) = |\mu|([0, T]) = \int_0^T |da(s)|.$$

In order to get the reverse inequality, it suffices to prove the second assertion. So we consider an increasing sequence $0 = t_0^n < t_1^n < \dots < t_{p_n}^n = T$ of subdivisions of $[0, T]$, whose mesh $\max\{t_i^n - t_{i-1}^n : 1 \leq i \leq p_n\}$ tends to 0. Although we are proving a "deterministic" result, we will use a martingale argument. Leaving aside the trivial case where $|\mu| = 0$, we introduce the probability space $\Omega = [0, T]$, which is equipped with the Borel σ -field $\mathcal{B} = \mathcal{B}([0, T])$ and the probability measure $P(ds) = (|\mu|([0, T]))^{-1} |\mu|(ds)$. On this probability space, we consider the discrete filtration $(\mathcal{B}_n)_{n \geq 0}$ such that, for every integer $n \geq 0$, $\mathcal{B}_n$ is the σ -field generated by the intervals $(t_{i-1}^n, t_i^n]$, $1 \leq i \leq p_n$.

We then set

$$X(s) = \mathbb{1}_{D_+}(s) - \mathbb{1}_{D_-}(s) = \frac{d\mu}{d|\mu|}(s),$$

and, for every $n \geq 0$,

$$X_n = \mathbb{E}[X \mid \mathcal{B}_n].$$

Properties of conditional expectation show that X_n is constant on every interval $(t_{i-1}^n, t_i^n]$ and takes the value

$$\frac{\mu((t_{i-1}^n, t_i^n])}{|\mu|((t_{i-1}^n, t_i^n])} = \frac{a(t_i^n) - a(t_{i-1}^n)}{|\mu|((t_{i-1}^n, t_i^n])}$$

on this interval. On the other hand, the sequence (X_n) is a closed martingale, with respect to the discrete filtration $(\mathcal{B}_n)$. Since X is measurable with respect to $\mathcal{B} = \bigvee_n \mathcal{B}_n$, this martingale converges to X in L^1 , by the convergence theorem for closed discrete martingales 1.3.16. In particular,

$$\lim_{n \rightarrow \infty} \mathbb{E}[|X_n|] = \mathbb{E}[|X|] = 1,$$

where the last equality is clear since $|X(s)| = 1$, $|\mu|(ds)$ a.e. The desired result follows by noting that

$$\mathbb{E}[|X_n|] = (|\mu|([0, T]))^{-1} \sum_{i=1}^{p_n} |a(t_i^n) - a(t_{i-1}^n)|,$$

and recalling that $|\mu|([0, T]) = \int_0^T |da(s)|$. ■

Remark In the usual presentation of functions with finite variation, one starts from the property that the supremum in the first display of the proposition is finite.

We now give a useful approximation lemma for the integral of a continuous function with respect to a function with finite variation

Lemma 1.5.3: Approximation for integral w.r.t finite variation function

If $f : [0, T] \rightarrow \mathbb{R}$ is a continuous function, and if $0 = t_0^n < t_1^n < \dots < t_{p_n}^n = T$ is a sequence of subdivisions of $[0, T]$ whose mesh tends to 0, we have

$$\int_0^T f(s) da(s) = \lim_{n \rightarrow \infty} \sum_{i=1}^{p_n} f(t_{i-1}^n) (a(t_i^n) - a(t_{i-1}^n)).$$

Proof for Lemma

Let f_n be defined on $[0, T]$ by $f_n(s) = f(t_{i-1}^n)$ if $s \in (t_{i-1}^n, t_i^n]$, $1 \leq i \leq p_n$, and $f_n(0) = f(0)$. Then,

$$\sum_{i=1}^{p_n} f(t_{i-1}^n) (a(t_i^n) - a(t_{i-1}^n)) = \int_{[0, T]} f_n(s) \mu(ds),$$

and the desired result follows by dominated convergence since $f_n(s) \rightarrow f(s)$ as $n \rightarrow \infty$, for every $s \in [0, T]$. ■

We say that a function $a : \mathbb{R}_+ \rightarrow \mathbb{R}$ is a *finite variation function on $\mathbb{R}_+$* if the restriction of a to $[0, T]$ has finite variation on $[0, T]$, for every $T > 0$. Then there is a unique σ -finite (positive) measure on $\mathbb{R}_+$ whose restriction to every interval $[0, T]$ is the total variation measure of the restriction of a to $[0, T]$, and we write

$$\int_0^\infty f(s) |da(s)|$$

for the integral of a nonnegative Borel function f on $\mathbb{R}_+$ with respect to this σ -finite measure. Furthermore, we can define

$$\int_0^\infty f(s) da(s) = \lim_{T \rightarrow \infty} \int_0^T f(s) da(s) \in (-\infty, \infty)$$

for any real Borel function f on $\mathbb{R}_+$ such that $\int_0^\infty |f(s)| |da(s)| < \infty$.

1.5.2 Finite Variation Processes

We now consider random variables and processes defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), P)$.

Definition 1.5.4: Finite Variation process

An adapted process $A = (A_t)_{t \geq 0}$ is called a *finite variation process* if all its sample paths are finite variation functions on $\mathbb{R}_+$. If in addition the sample paths are nondecreasing functions, the process A is called an *increasing process*.

Remark In particular, $A_0 = 0$ and the sample paths of A are continuous – one can define finite variation processes with càdlàg sample paths, but in this book we consider **only the case of continuous sample paths**. Our special convention that the initial value of a finite variation process is 0 will be convenient for certain uniqueness statements. If A is a finite variation process, the process

$$V_t = \int_0^t |dA_s|$$

is an increasing process. Indeed, it is clear that the sample paths of V are nondecreasing functions (as well as continuous functions that vanish at $t = 0$). The fact that V_t is an $\mathcal{F}_t$ -measurable random variable can be deduced from the second part of Proposition 1.5.2. Writing $A_t = \frac{1}{2}(V_t + A_t) - \frac{1}{2}(V_t - A_t)$ shows that any finite variation process can be written as the difference of two increasing processes (the converse is obvious).

Proposition 1.5.5

Let A be a finite variation process, and let H be a progressive process such that

$$\forall t \geq 0, \forall \omega \in \Omega, \int_0^t |H_s(\omega)| |dA_s(\omega)| < \infty.$$

Then the process $H \cdot A = ((H \cdot A)_t)_{t \geq 0}$ defined by

$$(H \cdot A)_t = \int_0^t H_s dA_s$$

is also a finite variation process.

Proof for Proposition.

By the observations preceding the statement of Proposition 1.5.2, we know that the sample paths of $H \cdot A$ are finite variation functions. It remains to verify that the process $H \cdot A$ is adapted. To this end, it is enough to check that, if $t > 0$ is fixed, if $h : \Omega \times [0, t] \rightarrow \mathbb{R}$ is measurable for the σ -field $\mathcal{F}_t \otimes \mathcal{B}([0, t])$, and if $\int_0^t |h(\omega, s)| |dA_s(\omega)| < \infty$ for every ω , then the variable $\int_0^t h(\omega, s) dA_s(\omega)$ is $\mathcal{F}_t$ -measurable.

If $h(\omega, s) = \mathbb{1}_{(u, v]}(s) \mathbb{1}_\Gamma(\omega)$ with $(u, v] \subset [0, t]$ and $\Gamma \in \mathcal{F}_t$, the result is immediate since $\int_0^t h(\omega, s) dA_s(\omega) = \mathbb{1}_\Gamma(\omega)(A_v(\omega) - A_u(\omega))$ in that case. A monotone class argument ?? then gives the case $h = \mathbb{1}_G$, $G \in \mathcal{F}_t \otimes \mathcal{B}([0, t])$. Finally, in the general case, we observe that we can write h as a pointwise limit of a sequence of simple functions (i.e., finite linear combinations of indicator functions of measurable sets) h_n such that $|h_n| \leq |h|$ for every n , and that we then have $\int_0^t h_n(\omega, s) dA_s(\omega) \rightarrow \int_0^t h(\omega, s) dA_s(\omega)$ by dominated convergence, for every $\omega \in \Omega$. $\square$

Remarks

- (i) It happens frequently that instead of the assumption of the proposition we have the weaker assumption

$$\text{a.s. } \forall t \geq 0, \int_0^t |H_s(\omega)| |dA_s(\omega)| < \infty.$$

If the filtration is complete, we can still define $H \cdot A$ as a finite variation process under this weaker assumption. We replace H by the process H' defined by

$$H'_t(\omega) = \begin{cases} H_t(\omega) & \text{if } \int_0^t |H_s(\omega)| |dA_s(\omega)| < \infty, \forall t, \\ 0 & \text{otherwise.} \end{cases}$$

Thanks to the fact that the filtration is complete, the process H' is still progressive, which allows us to define $H \cdot A = H' \cdot A$. We will use this extension of Proposition 1.5.5 implicitly in what follows.

- (ii) Under appropriate assumptions (if H and K are progressive and $\int_0^t |H_s| |dA_s| < \infty$,

$\int_0^t |H_s K_s| |dA_s| < \infty$ for every $t \geq 0$), we have the *associativity property*

$$K \cdot (H \cdot A) = (KH) \cdot A.$$

This indeed follows from the analogous deterministic result saying informally that $k(s)(h(s) \mu(ds)) = (k(s)h(s)) \mu(ds)$ if $h(s)$ and $k(s)h(s)$ are integrable with respect to the signed measure μ on $[0, t]$.

An important special case of the proposition is the case where $A_t = t$. If H is a progressive process such that

$$\forall t \geq 0, \forall \omega \in \Omega, \int_0^t |H_s(\omega)| ds < \infty,$$

the process $\int_0^t H_s ds$ is a finite variation process.

1.6 Continuous Local Martingales

We consider again a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), P)$. If T is a stopping time, and if $X = (X_t)_{t \geq 0}$ is an adapted process with continuous sample paths, we will write X^T for process X stopped at T , defined by $X_t^T = X_{t \wedge T}$ for every $t \geq 0$. It is useful to observe that, if S is another stopping time,

$$(X^T)^S = (X^S)^T = X^{S \wedge T}.$$

1.6.1 Definition and Basic Properties of Continuous Local Martingales

Definition 1.6.1: continuous local martingale

An adapted process $M = (M_t)_{t \geq 0}$ with continuous sample paths and such that $M_0 = 0$ a.s. is called a *continuous local martingale* if there exists a nondecreasing sequence $(T_n)_{n \geq 0}$ of stopping times such that $T_n \uparrow \infty$ (i.e. $T_n(\omega) \uparrow \infty$ for every ω) and, for every n , the stopped process M^{T_n} is a uniformly integrable martingale.

More generally, when we do not assume that $M_0 = 0$ a.s., we say that M is a continuous local martingale if the process $N_t = M_t - M_0$ is a continuous local martingale. In all cases, we say that the sequence of stopping times (T_n) *reduces* M if $T_n \uparrow \infty$ and, for every n , the stopped process M^{T_n} is a uniformly integrable martingale. **Remarks**

1. We do not require in the definition of a continuous local martingale that the variables M_t are in L^1 (compare with the definition of martingales). In particular, the variable M_0 may be any $\mathcal{F}_0$ -measurable random variable.
2. Any martingale with continuous sample paths is a continuous local martingale (see property (a) below) but the converse is false, and for this reason we will sometimes speak of *true martingales* to emphasize the difference with local martingales.

3. One can define a notion of local martingale with càdlàg sample paths. In this course, however, we consider only continuous local martingales.

The following properties are easily established.

Proposition 1.6.2

1. A martingale with continuous sample paths is a continuous local martingale, and the sequence $T_n = n$ reduces M .
2. In the definition of a continuous local martingale starting from 0, one can replace uniformly integrable martingale by *martingale* (indeed, one can then observe that $M^{T_n \wedge n}$ is uniformly integrable, and we still have $T_n \wedge n \uparrow \infty$).
3. If M is a continuous local martingale, then, for every stopping time T , M^T is a continuous local martingale (this follows from Corollary of Optional Stopping Theorem 1.4.1).
4. If (T_n) reduces M and if (S_n) is a sequence of stopping times such that $S_n \uparrow \infty$, then the sequence $(T_n \wedge S_n)$ also reduces M (use same Corollary again).
5. The space of all continuous local martingales is a vector space (to check stability under addition, note that if M and M' are two continuous local martingales such that $M_0 = 0$ and $M'_0 = 0$, if the sequence (T_n) reduces M and if the sequence (T'_n) reduces M' , property (d) shows that the sequence $T_n \wedge T'_n$ reduces $M + M'$).

The next proposition gives three other useful properties of local martingales.

Proposition 1.6.3

1. A *nonnegative continuous local martingale* M such that $M_0 \in L^1$ is a *supermartingale*.
2. A *continuous local martingale* M such that there exists a random variable $Z \in L^1$ with $|M_t| \leq Z$ for every $t \geq 0$ (in particular a *bounded continuous local martingale*) is a *uniformly integrable martingale*.
3. If M is a *continuous local martingale* and $M_0 = 0$ (or more generally $M_0 \in L^1$), the sequence of stopping times

$$T_n = \inf\{t \geq 0 : |M_t| \geq n\}$$

reduces M .

Theorem 1.6.4: Total variation of Local Martingales

Let M be a continuous local martingale. Assume that M is also a finite variation process (in particular $M_0 = 0$). Then $M_t = 0$ for every $t \geq 0$, a.s.

Proof for Theorem.

Set

$$\tau_n = \inf \left\{ t \geq 0 : \int_0^t |dM_s| \geq n \right\}$$

for every integer $n \geq 0$. By Definition 1.2.10, τ_n is a stopping time (recall that $\int_0^t |dM_s|$ is an increasing process if M is a finite variation process).

Fix $n \geq 0$ and set $N = M^{\tau_n}$. Note that, for every $t \geq 0$,

$$|N_t| = |M_{t \wedge \tau_n}| \leq \int_0^t \wedge \tau_n |dM_s| \leq n.$$

By Proposition 1.6.3, N is a (bounded) martingale. Let $t > 0$ and let $0 = t_0 < t_1 < \dots < t_p = t$ be any subdivision of $[0, t]$. Then, from Proposition 1.2.3, we have

$$\begin{aligned} E[N_t^2] &= \sum_{i=1}^p E[(N_{t_i} - N_{t_{i-1}})^2] \\ &\leq E \left[\left(\sup_{1 \leq i \leq p} |N_{t_i} - N_{t_{i-1}}| \right) \sum_{i=1}^p |N_{t_i} - N_{t_{i-1}}| \right] \\ &\leq n E \left[\sup_{1 \leq i \leq p} |N_{t_i} - N_{t_{i-1}}| \right] \end{aligned}$$

noting that $\int_0^t |dN_s| \leq n$ by the definition of τ_n , and using Proposition 1.5.2.

We now apply the preceding bound to a sequence $0 = t_0^k < t_1^k < \dots < t_{p_k}^k = t$ of subdivisions of $[0, t]$ whose mesh tends to 0. Using the continuity of sample paths, and the fact that N is bounded (to justify dominated convergence), we get

$$\lim_{k \rightarrow \infty} E \left[\sup_{1 \leq i \leq p_k} |N_{t_i^k} - N_{t_{i-1}^k}| \right] = 0.$$

We then conclude that $E[N_t^2] = 0$, hence $M_{t \wedge \tau_n} = 0$ a.s. Letting n tend to ∞ , we get that $M_t = 0$ a.s. ■

1.6.2 The Quadratic Variation of a Continuous Local Martingale

Theorem 1.6.5: Quadratic variation existence

Let $M = (M_t)_{t \geq 0}$ be a continuous local martingale. There exists an increasing process denoted by $(\langle M, M \rangle_t)_{t \geq 0}$, which is unique up to indistinguishability, such that $M_t^2 - \langle M, M \rangle_t$ is a continuous local martingale. Furthermore, for every fixed $t > 0$, if $0 = t_0^n < t_1^n < \dots < t_{p_n}^n = t$ is an increasing sequence of subdivisions of $[0, t]$ with mesh tending to 0, we have :

$$\langle M, M \rangle_t = \lim_{n \rightarrow \infty} \sum_{i=1}^{p_n} (M_{t_i^n} - M_{t_{i-1}^n})^2$$

In probability. The process $\langle M, M \rangle$ is called the quadratic variation of M .

Proof for Theorem.

The proof can be found in [Stochastic] ■

Remarks :

1. We observe that the process $\langle M, M \rangle$ does not depend on the initial value M_0 , but only on the increments of M : if $M_t = M_0 + N_t$, we have $\langle M, M \rangle = \langle N, N \rangle$. This is obvious from the second assertion of the theorem, and this will also be clear from the proof that follows.
2. In the second assertion of the theorem, it is in fact not necessary to assume that the sequence of subdivisions is increasing.

Now we present some properties of the quadratic variation :

Proposition 1.6.6

Let M be a continuous local martingale and let T be a stopping time.

1. We have a.s. for every $t \geq 0$,

$$\langle M^T, M^T \rangle_t = \langle M, M \rangle_{t \wedge T}.$$

2. If $M_0 = 0$. Then we have $\langle M, M \rangle = 0$ if and only if $M = 0$.

Proof for Proposition.

1. This follows from the fact that $M_{t \wedge T}^2 - \langle M, M \rangle_{t \wedge T}$ is a continuous local martingale (cf. property (3) of continuous local martingales in Proposition 1.6.2).
2. Suppose that $\langle M, M \rangle = 0$. Then M_t^2 is a nonnegative continuous local martingale and, by Proposition 1.6.3 (1), M_t^2 is a supermartingale, hence $E[M_t^2] \leq E[M_0^2] = 0$, so that $M_t = 0$ for every t . The converse is obvious.

The next theorem shows that properties of a continuous local martingale are closely related to those of its quadratic variation. If A is an increasing process, A_∞ denotes the increasing limit of A_t as $t \rightarrow \infty$ (this limit always exists in $[0, \infty]$).

Theorem 1.6.7

Let M be a continuous local martingale with $M_0 \in L^2$.

1. The following are equivalent :

- (a) M is a (true) martingale bounded in L^2 .
- (b) $E[\langle M, M \rangle_\infty] < \infty$.

Furthermore, if these properties hold, the process $M_t^2 - \langle M, M \rangle_t$ is a uniformly integrable martingale, and in particular $E[M_\infty^2] = E[M_0^2] + E[\langle M, M \rangle_\infty]$.

2. The following are equivalent :

- (a) M is a (true) martingale and $M_t \in L^2$ for every $t \geq 0$.
- (b) $E[\langle M, M \rangle_t] < \infty$ for every $t \geq 0$.

Furthermore, if these properties hold, the process $M_t^2 - \langle M, M \rangle_t$ is a martingale.

Proof for Theorem.

Let's prove (1), Replacing M by $M - M_0$, we may assume that $M_0 = 0$ in the proof. Let us first assume that M is a martingale bounded in L^2 . Doob's inequality in L^2 (Proposition 1.3.5 (4)) shows that, for every $T > 0$,

$$E \left[\sup_{0 \leq t \leq T} M_t^2 \right] \leq 4E[M_T^2].$$

By letting T go to ∞ , we have

$$E \left[\sup_{t \geq 0} M_t^2 \right] \leq 4 \sup_{t \geq 0} E[M_t^2] =: C < \infty.$$

Set $S_n = \inf\{t \geq 0 : \langle M, M \rangle_t \geq n\}$. Then the continuous local martingale $M_{t \wedge S_n}^2 - \langle M, M \rangle_{t \wedge S_n}$ is dominated by the variable

$$\sup_{s \geq 0} M_s^2 + n,$$

which is integrable. From Proposition 1.6.3 (2), we get that this continuous local martingale is a uniformly integrable martingale, hence

$$E[\langle M, M \rangle_{t \wedge S_n}] = E[M_{t \wedge S_n}^2] \leq E \left[\sup_{s \geq 0} M_s^2 \right] \leq C.$$

By letting n , and then t tend to infinity, and using monotone convergence, we get $E[\langle M, M \rangle_\infty] \leq C < \infty$.

Conversely, assume that $E[\langle M, M \rangle_\infty] < \infty$. Set $T_n = \inf\{t \geq 0 : |M_t| \geq n\}$. Then the continuous local martingale $M_{t \wedge T_n}^2 - \langle M, M \rangle_{t \wedge T_n}$ is dominated by the variable

$$n^2 + \langle M, M \rangle_\infty,$$

which is integrable. From Proposition 1.6.3 (2) again, this continuous local martingale is a uniformly integrable martingale, hence, for every $t \geq 0$,

$$E[M_{t \wedge T_n}^2] = E[\langle M, M \rangle_{t \wedge T_n}] \leq E[\langle M, M \rangle_\infty] =: C' < \infty.$$

By letting $n \rightarrow \infty$ and using Fatou's lemma, we get $E[M_t^2] \leq C'$, so that the collection $(M_t)_{t \geq 0}$ is bounded in L^2 . We have not yet verified that $(M_t)_{t \geq 0}$ is a martingale. However, the previous bound on $E[M_{t \wedge T_n}^2]$ shows that the sequence $(M_{t \wedge T_n})_{n \geq 1}$ is uniformly integrable, and therefore converges both a.s. and in L^1 to M_t , for every $t \geq 0$. Recalling that M^{T_n} is a martingale (Proposition 1.6.3 (3)), the L^1 -convergence allows us to pass to the limit $n \rightarrow \infty$ in the martingale property $E[M_{t \wedge T_n} | \mathcal{F}_s] = M_{s \wedge T_n}$, for $0 \leq s < t$, and to get that M is a martingale.

Finally, if properties (a) and (b) hold, the continuous local martingale $M^2 - \langle M, M \rangle$

is dominated by the integrable variable

$$\sup_{t \geq 0} M_t^2 + \langle M, M \rangle_\infty$$

and is therefore (by Proposition 1.6.3 (2)) a uniformly integrable martingale. ■

Remark :

1. In property (a) of (1) (or of (2)), it is essential to suppose that M is a martingale, and not only a continuous local martingale. Doob's inequality used in the previous proof is not valid in general for a continuous local martingale!
2. It suffices to apply (1) to $(M_{t \wedge a})_{t \geq 0}$ for every choice of $a \geq 0$, to prove (2).

1.6.3 The Bracket of Two Continuous Local Martingales

Definition 1.6.8: Bracket of two continuous local martingales

If M and N are two continuous local martingales, the *bracket* $\langle M, N \rangle$ is the finite variation process defined by setting, for every $t \geq 0$,

$$\langle M, N \rangle_t = \frac{1}{2}(\langle M + N, M + N \rangle_t - \langle M, M \rangle_t - \langle N, N \rangle_t).$$

Let us state a few easy properties of the bracket.

Proposition 1.6.9

1. $\langle M, N \rangle$ is the unique (up to indistinguishability) finite variation process such that $M_t N_t - \langle M, N \rangle_t$ is a continuous local martingale.
2. The mapping $(M, N) \mapsto \langle M, N \rangle$ is bilinear and symmetric.
3. If $0 = t_0^n < t_1^n < \dots < t_{p_n}^n = t$ is an increasing sequence of subdivisions of $[0, t]$ with mesh tending to 0, we have

$$\lim_{n \rightarrow \infty} \sum_{i=1}^{p_n} (M_{t_i^n} - M_{t_{i-1}^n})(N_{t_i^n} - N_{t_{i-1}^n}) = \langle M, N \rangle_t$$

in probability.

4. For every stopping time T , $\langle M^T, N^T \rangle_t = \langle M^T, N \rangle_t = \langle M, N \rangle_{t \wedge T}$.
5. If M and N are two martingales (with continuous sample paths) bounded in L^2 , $M_t N_t - \langle M, N \rangle_t$ is a uniformly integrable martingale. Consequently, $\langle M, N \rangle_\infty$ is well defined as the almost sure limit of $\langle M, N \rangle_t$ as $t \rightarrow \infty$, is integrable, and satisfies

$$\mathbb{E}[M_\infty N_\infty] = \mathbb{E}[M_0 N_0] + \mathbb{E}[\langle M, N \rangle_\infty].$$

Remark : A consequence of (iv) is the fact that $M^T(N - N^T)$ is a continuous local

martingale, which is not so easy to prove directly.

Definition 1.6.10: Orthogonal continuous local martingales

Two continuous local martingales M and N are said to be orthogonal if $\langle M, N \rangle = 0$, which holds if and only if MN is a continuous local martingale.

If M and N are two orthogonal martingales bounded in L^2 , we have $E[M_t N_t] = E[M_0 N_0]$, and even $E[M_S N_S] = E[M_0 N_0]$ for any stopping time S . This follows from Theorem 1.4.1, using property (5) of Proposition 1.6.9.

Theorem 1.6.11: Kunita-Watanabe inequality

Let M and N be two continuous local martingales and let H and K be two measurable processes. Then, a.s.,

$$\int_0^\infty |H_s| |K_s| |d\langle M, N \rangle_s| \leq \left(\int_0^\infty H_s^2 d\langle M, M \rangle_s \right)^{1/2} \left(\int_0^\infty K_s^2 d\langle N, N \rangle_s \right)^{1/2}.$$

Proof for Theorem.

Only in this proof, we use the special notation $\langle M, N \rangle_s^t = \langle M, N \rangle_t - \langle M, N \rangle_s$ for $0 \leq s \leq t$. The first step of the proof is to observe that we have a.s. for every choice of the rationals $s < t$ (and also by continuity for every reals $s < t$),

$$|\langle M, N \rangle_s^t| \leq \sqrt{\langle M, M \rangle_s^t} \sqrt{\langle N, N \rangle_s^t}.$$

Indeed, this follows from the approximations of $\langle M, M \rangle$ and $\langle M, N \rangle$ given in Theorem 1.6.5 and in Proposition 1.6.9 respectively (note that these approximations are easily extended to the increments of $\langle M, M \rangle$ and $\langle M, N \rangle$), together with the Cauchy-Schwarz inequality. From now on, we fix ω such that the inequality of the last display holds for every $s < t$, and we argue with this value of ω (the remaining part of the argument is *deterministic*).

We then observe that we also have, for every $0 \leq s \leq t$,

$$\int_s^t |d\langle M, N \rangle_u| \leq \sqrt{\langle M, M \rangle_s^t} \sqrt{\langle N, N \rangle_s^t}. \quad (*)$$

Indeed, we use Proposition 1.5.2, noting that, for any subdivision $s = t_0 < t_1 < \dots < t_p = t$, we can bound

$$\begin{aligned} \sum_{i=1}^p |\langle M, N \rangle_{t_{i-1}}^{t_i}| &\leq \sum_{i=1}^p \sqrt{\langle M, M \rangle_{t_{i-1}}^{t_i}} \sqrt{\langle N, N \rangle_{t_{i-1}}^{t_i}} \\ &\leq \left(\sum_{i=1}^p \langle M, M \rangle_{t_{i-1}}^{t_i} \right)^{1/2} \left(\sum_{i=1}^p \langle N, N \rangle_{t_{i-1}}^{t_i} \right)^{1/2} \\ &= \sqrt{\langle M, M \rangle_s^t} \sqrt{\langle N, N \rangle_s^t}. \end{aligned}$$

We then get that, for every bounded Borel subset A of $\mathbb{R}_+$,

$$\int_A |\mathrm{d}\langle M, N \rangle_u| \leq \sqrt{\int_A \mathrm{d}\langle M, M \rangle_u} \sqrt{\int_A \mathrm{d}\langle N, N \rangle_u}.$$

When $A = [s, t]$, this is the bound (*). If A is a finite union of intervals, this follows from (*) and another application of the Cauchy-Schwarz inequality. A monotone class argument shows that the inequality of the last display remains valid for any bounded Borel set A (Theorem ??).

Next let $h = \sum_{i=1}^p \lambda_i \mathbf{1}_{A_i}$ and $k = \sum_{i=1}^p \mu_i \mathbf{1}_{A_i}$ be two nonnegative simple functions on $\mathbb{R}_+$ with bounded support contained in $[0, K]$, for some $K > 0$. Here $A_1, \dots, A_p$ is a measurable partition of $[0, K]$, and $\lambda_1, \dots, \lambda_p, \mu_1, \dots, \mu_p$ are reals (we can always assume that h and k are expressed in terms of the same partition). Then,

$$\begin{aligned} \int h(s)k(s)|\mathrm{d}\langle M, N \rangle_s| &= \sum_{i=1}^p \lambda_i \mu_i \int_{A_i} |\mathrm{d}\langle M, N \rangle_s| \\ &\leq \sum_{i=1}^p \lambda_i \mu_i \sqrt{\int_{A_i} \mathrm{d}\langle M, M \rangle_s} \sqrt{\int_{A_i} \mathrm{d}\langle N, N \rangle_s} \\ &\leq \left(\sum_{i=1}^p \lambda_i^2 \int_{A_i} \mathrm{d}\langle M, M \rangle_s \right)^{1/2} \left(\sum_{i=1}^p \mu_i^2 \int_{A_i} \mathrm{d}\langle N, N \rangle_s \right)^{1/2} \\ &= \left(\int h(s)^2 \mathrm{d}\langle M, M \rangle_s \right)^{1/2} \left(\int k(s)^2 \mathrm{d}\langle N, N \rangle_s \right)^{1/2}, \end{aligned}$$

which gives the desired inequality for simple functions. Since every nonnegative Borel function is a monotone increasing limit of simple functions with bounded support, an application of the monotone convergence theorem completes the proof. $\blacksquare$

1.6.4 The Hilbert Space $\mathbb{H}^2$

We write $\mathbb{H}^2$ for the space of all continuous martingales M which are bounded in L^2 and such that $M_0 = 0$, with the usual convention that two indistinguishable processes are identified. Equivalently, $M \in \mathbb{H}^2$ if and only if M is a continuous local martingale such that $M_0 = 0$ and

$$\mathbb{E}[\langle M, M \rangle_\infty] < \infty;$$

see Theorem 1.6.7. By Theorem 1.3.16, every $M \in \mathbb{H}^2$ admits an almost sure and L^2 limit $M_\infty \in L^2$, and

$$M_t = \mathbb{E}[M_\infty \mid \mathcal{F}_t], \quad t \geq 0.$$

If $M, N \in \mathbb{H}^2$, Proposition 1.6.9 (v) shows that $\langle M, N \rangle_\infty$ is well defined and integrable. We may therefore define a symmetric bilinear form on $\mathbb{H}^2$ by

$$(M, N)_{\mathbb{H}^2} = \mathbb{E}[\langle M, N \rangle_\infty] = \mathbb{E}[M_\infty N_\infty].$$

The second equality follows again from Proposition 1.6.9 (v), since $M_0 = N_0 = 0$. In

particular, $(M, M)_{\mathbb{H}^2} = 0$ if and only if $M = 0$. The associated norm is

$$\|M\|_{\mathbb{H}^2} = \left((M, M)_{\mathbb{H}^2} \right)^{1/2} = (\mathbb{E}[\langle M, M \rangle_{\infty}])^{1/2} = (\mathbb{E}[M_{\infty}^2])^{1/2}.$$

Proposition 1.6.12

The space $\mathbb{H}^2$, equipped with the scalar product $(\cdot, \cdot)_{\mathbb{H}^2}$, is a Hilbert space.

Proof for Proposition.

Let $(M^n)_{n \geq 1}$ be a Cauchy sequence in $\mathbb{H}^2$. Then $(M_{\infty}^n)_{n \geq 1}$ is Cauchy in L^2 , hence converges in L^2 to some $X \in L^2$. For each $t \geq 0$, the sequence $(M_t^n)_{n \geq 1}$ converges in L^2 to $M_t := \mathbb{E}[X \mid \mathcal{F}_t]$. Since $\mathbb{E}[M_{\infty}^n \mid \mathcal{F}_0] = M_0^n = 0$ for every n , we also have $M_0 = \mathbb{E}[X \mid \mathcal{F}_0] = 0$. Thus $(M_t)_{t \geq 0}$ is a martingale bounded in L^2 .

Moreover, Doob's L^2 inequality gives, for every $n, m \geq 1$,

$$\mathbb{E} \left[\sup_{t \geq 0} |M_t^n - M_t^m|^2 \right] \leq 4 \mathbb{E} [|M_{\infty}^n - M_{\infty}^m|^2].$$

By extracting a subsequence, we obtain almost sure uniform convergence of M^n to M on $\mathbb{R}_+$. Since each M^n has continuous sample paths, M has a continuous modification. Hence $M \in \mathbb{H}^2$. Finally,

$$\|M^n - M\|_{\mathbb{H}^2}^2 = \mathbb{E} [|M_{\infty}^n - X|^2] \longrightarrow 0,$$

which proves that $\mathbb{H}^2$ is complete. ■

1.7 Continuous Semimartingales

We now introduce the class of processes for which we will develop the theory of stochastic integrals.

Definition 1.7.1: Continuous semimartingale

A process $X = (X_t)_{t \geq 0}$ is a *continuous semimartingale* if it can be written in the form

$$X_t = M_t + A_t,$$

where M is a continuous local martingale and A is a finite variation process.

The decomposition $X = M + A$ is then unique up to indistinguishability thanks to Theorem 1.6.4. We say that this is the *canonical decomposition* of X .

By construction, continuous semimartingales have continuous sample paths. It is possible to define a notion of semimartingale with càdlàg sample paths, but in this book, we will only deal with *continuous* semimartingales, and for this reason we sometimes omit the word continuous.

Definition 1.7.2: Bracket of semimartingales

Let $X = M + A$ and $Y = M' + A'$ be the canonical decompositions of two continuous semimartingales X and Y . The *bracket* $\langle X, Y \rangle$ is the finite variation process defined by

$$\langle X, Y \rangle_t = \langle M, M' \rangle_t.$$

In particular, we have $\langle X, X \rangle_t = \langle M, M \rangle_t$.

Proposition 1.7.3

Let $0 = t_0^n < t_1^n < \dots < t_{p_n}^n = t$ be an increasing sequence of subdivisions of $[0, t]$ whose mesh tends to 0. Then,

$$\lim_{n \rightarrow \infty} \sum_{i=1}^{p_n} (X_{t_i^n} - X_{t_{i-1}^n})(Y_{t_i^n} - Y_{t_{i-1}^n}) = \langle X, Y \rangle_t$$

in probability.

Proof for Proposition.

We treat the case where $X = Y$ and leave the general case to the reader. We have

$$\sum_{i=1}^{p_n} (X_{t_i^n} - X_{t_{i-1}^n})^2 = \sum_{i=1}^{p_n} (M_{t_i^n} - M_{t_{i-1}^n})^2 + \sum_{i=1}^{p_n} (A_{t_i^n} - A_{t_{i-1}^n})^2 + 2 \sum_{i=1}^{p_n} (M_{t_i^n} - M_{t_{i-1}^n})(A_{t_i^n} - A_{t_{i-1}^n}).$$

By Theorem 1.6.5,

$$\lim_{n \rightarrow \infty} \sum_{i=1}^{p_n} (M_{t_i^n} - M_{t_{i-1}^n})^2 = \langle M, M \rangle_t = \langle X, X \rangle_t,$$

in probability. On the other hand,

$$\begin{aligned} \sum_{i=1}^{p_n} (A_{t_i^n} - A_{t_{i-1}^n})^2 &\leq \left(\sup_{1 \leq i \leq p_n} |A_{t_i^n} - A_{t_{i-1}^n}| \right) \sum_{i=1}^{p_n} |A_{t_i^n} - A_{t_{i-1}^n}| \\ &\leq \left(\int_0^t |dA_s| \right) \sup_{1 \leq i \leq p_n} |A_{t_i^n} - A_{t_{i-1}^n}|, \end{aligned}$$

which tends to 0 a.s. when $n \rightarrow \infty$ by the continuity of sample paths of A . The same argument shows that

$$\left| \sum_{i=1}^{p_n} (A_{t_i^n} - A_{t_{i-1}^n})(M_{t_i^n} - M_{t_{i-1}^n}) \right| \leq \left(\int_0^t |dA_s| \right) \sup_{1 \leq i \leq p_n} |M_{t_i^n} - M_{t_{i-1}^n}|$$

tends to 0 a.s. ■